PART-1 : PHYSICS

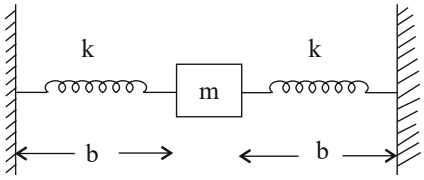
SECTION–I(i) : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases

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1. In the adjacent figure, a block of mass m is rigidly attached to two identical spring of stiffness K each. The other ends of the spring are connected to the fixed wall. When the block is in equilibrium, length of each spring is b, which is greater than the natural length l of the spring. The Time – period of the oscillation of the block if it is displaced by small distance X perpendicular to the length of the spring and released. (neglect gravity)

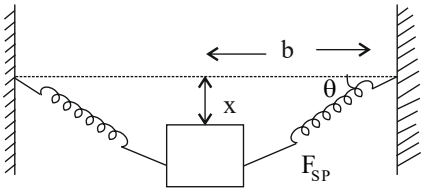


(A) $2\pi\sqrt{\dfrac{mb}{k(b-l)}}$ (B) $2\pi\sqrt{\dfrac{mb}{2k(b-l)}}$

(C) $2\pi\sqrt{\dfrac{m(b-l)}{kb}}$ (D) $2\pi\sqrt{\dfrac{m(b-l)}{2kb}}$

Ans. B

Sol.



when the block is displaced by x, the extension in the spring is equal to $\sqrt{b^2+x^2}-l$

$\therefore$ spring force $=k\left(\sqrt{b^2+x^2}-l\right)$

Net restoring force towards the equilibrium position is $2F_{SP}\sin\theta$

or $m\dfrac{d^2x}{dt^2}=-2k\left(\sqrt{b^2+x^2}-l\right)\left(\dfrac{x}{\sqrt{b^2+x^2}}\right)$

$$m\frac{d^2x}{dt^2}=-2k\left[1-\frac{l}{b\sqrt{1+\dfrac{x^2}{b^2}}}\right]x$$

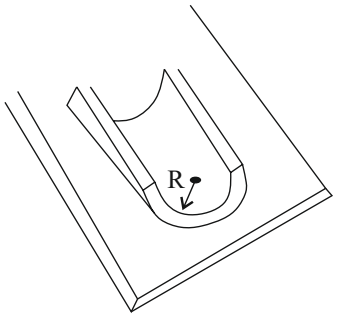
$$m\frac{d^2x}{dt^2} = -2k\left[\frac{b-l}{b}\right]x \qquad \{\text{for small displacement}\}$$

$$\Rightarrow \frac{d^2x}{dt^2} = \frac{-2k(b-l)}{mb}x$$

Therefore, block will perform SHM with its angular frequency $\omega = \sqrt{\frac{2k(b-l)}{mb}}$ and time period

$$T = 2\pi\sqrt{\frac{mb}{2k(b-l)}}$$

2. The semi-circular cylindrical shell of radius R with small but uniform wall thickness is set into small oscillation on the horizontal surface. If no slipping occurs, then the expression for the period T of each complete oscillation is :

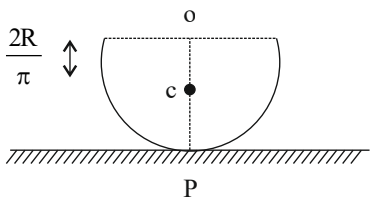


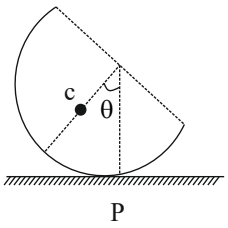
(A) $T = 2\pi\sqrt{\frac{(\pi-2)R}{g}}$ (B) $T = 2\pi\sqrt{\frac{(\pi-4)R}{g}}$

(C) $T = 2\pi\sqrt{\frac{2R}{g}}$ (D) $T = 2\pi\sqrt{\frac{(\pi-6)R}{g}}$

Ans. A

Sol.





writing energy equation at any instant

$$E = mg\left(\frac{2R}{\pi}\right)(1-\cos\theta) + \frac{1}{2}I\omega^2$$

where, I = Moment of inertia about P

$$I_{cm} = mR^2 - m\left(\frac{2R}{\pi}\right)^2$$

$$\therefore I_p = I_{cm} + m\left(R - \frac{2R}{\pi}\right)^2$$

$$= mR^2 - m\left(\frac{2R}{\pi}\right)^2 + m\left(R - \frac{2R}{\pi}\right)^2$$

$$= mR^2 - m\left(\frac{2R}{\pi}\right)^2 + mR^2 + m\left(\frac{2R}{\pi}\right)^2 - 2mR\left(\frac{2R}{\pi}\right)$$

$$= 2mR^2 - \frac{4mR^2}{\pi}$$

$$\Rightarrow E = mg\left(\frac{2R}{\pi}\right)(1-\cos\theta) + \frac{1}{2}\times 2\left(mR^2 - \frac{2mR^2}{\pi}\right)\omega^2$$

Since, total energy is constant. Differentiating it we get

$$0 = mg\left(\frac{2R}{\pi}\right)\sin\theta \times \frac{d\theta}{dt} + \left(mR^2 - \frac{2mR^2}{\pi}\right).2\omega\frac{d\omega}{dt}$$

$$0 = mg\left(\frac{2R}{\pi}\right)\sin\theta + 2\left(mR^2 - \frac{2mR^2}{\pi}\right)\frac{d^2\theta}{dt^2}$$

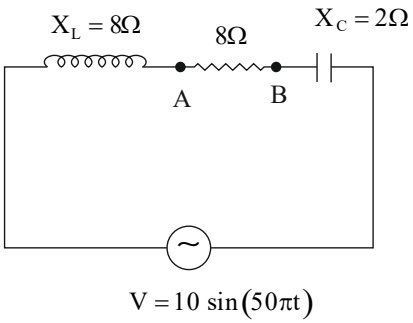
for small angle

$$2mR^2\left(1-\frac{2}{\pi}\right)\frac{d^2\theta}{dt^2} = -mg\left(\frac{2R}{\pi}\right)\theta$$

or $$\frac{d^2\theta}{dt^2} = \frac{-g}{R(\pi-2)}\theta$$

$$\omega^2 = \frac{g}{R(\pi-2)} \Rightarrow \omega = \sqrt{\frac{g}{R(\pi-2)}} \text{ and } T = 2\pi\sqrt{\frac{(\pi-2)R}{g}}$$

3. The instantaneous potential difference between points A and B is :



(A) $8\sin\left(50\pi t + 37\frac{\pi}{180}\right)$

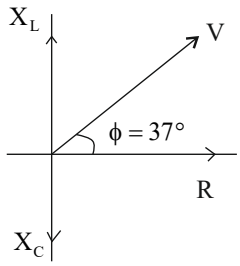
(B) $8\sin\left(50\pi t - 37\frac{\pi}{180}\right)$

(C) $10\sin(50\pi t)$

(D) $10\cos(50\pi t)$

Ans. B

Sol.



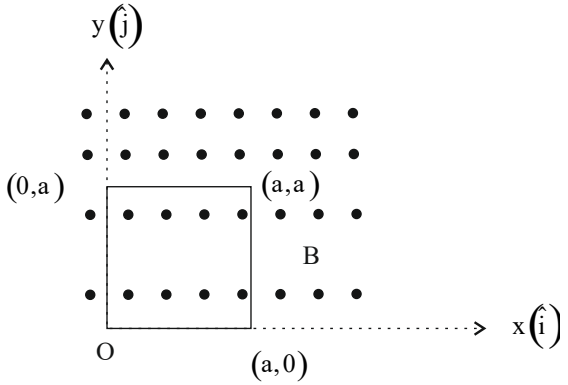
Impedance $z=\sqrt{(8-2)^2+(8)^2}=10\Omega$

current lags voltage by 37°, then

$i=\frac{10}{10}\sin(50\pi t-37°)$

$V_{AB}=i\times R=8\sin(50\pi t-37°)$

4. A square loop of side 'a' is placed in x - y plane as shown in figure. In this region there is non-uniform time dependent magnetic field $\vec{B}=(cy^3t^2)\hat{k}$. [where t is time and c is constant] then magnitude of emf induced in loop is



(A) Zero

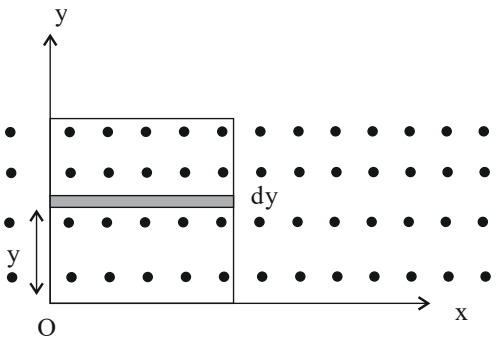
(B) $\frac{ca^4t^2}{5}$

(C) $\frac{ca^5t}{2}$

(D) Data is insufficient

Ans. C

Sol.



$\phi=\int\vec{B}.\overrightarrow{dA}$

$$\phi = \int cy^3t^2 ady$$

$$\phi = ct^2a.\int_0^a y^3dy$$

$$\phi = ct^2a\frac{a^4}{4}$$

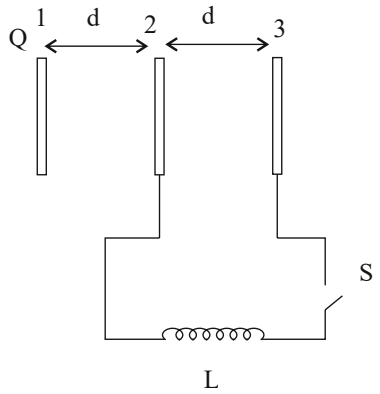
$$\phi = \frac{ct^2a^5}{4}$$

$$e = \left|-\frac{d\varphi}{dt}\right|$$

Now induced e.m.f

$$e = \left|-\frac{2cta^5}{4}\right| = \frac{2cta^5}{4} = \frac{cta^5}{2}$$

5. Three identical large plates are fixed at separation of d from each other as shown. The area of each plate is A. Plate 1 is given charge +Q while plates 2 and 3 are neutral and are connected to each other through coil of inductances L and switch S. If resistance of all connected wires is neglected the maximum current flow through coil after closing switch is $\left(C = \varepsilon_0 A / d\right)$ (neglect fringe effect)

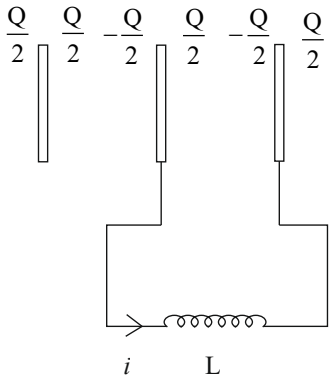


(A) $\frac{Q_0}{\sqrt{LC}}$ (B) $\frac{Q_0}{\sqrt{2LC}}$

(C) $\frac{2Q_0}{\sqrt{LC}}$ (D) $\frac{Q_0}{2\sqrt{LC}}$

Ans. D

Sol.



$$\frac{\left(\frac{Q}{2}\right)^2}{2C} = \frac{1}{2}Li_0^2$$

$$\Rightarrow I_0 = \frac{Q}{2\sqrt{LC}}$$

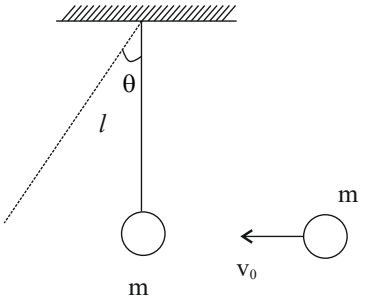
SECTION–I(ii) : (Maximum Marks : 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

6. A simple pendulum consists of a bob of mass m and a light string of length l as shown in the figure. Another identical ball moving with the small velocity v_0 collides with the pendulum's bob and sticks to it. For this new pendulum of mass 2m, mark out the correct statement(s)



(A) Time period of the pendulum is $2\pi\sqrt{\frac{l}{g}}$

(B) The equation of motion for this pendulum is $\theta = \frac{v_0}{2\sqrt{gl}}\sin\left[\sqrt{\frac{g}{l}}t\right]$

(C) The equation of motion for this pendulum is $\theta = \frac{v_0}{2\sqrt{gl}}\cos\left[\sqrt{\frac{g}{l}}t\right]$

(D) Time period of the pendulum is $2\pi\sqrt{\frac{2l}{g}}$

Ans. AB

Sol. The time period of simple pendulum is independent of mass, so it would be same as that $T = 2\pi\sqrt{l/g}$

After collision, the combined mass acquires a velocity of $v_0/2$, as a result of this velocity, the mass (2m) moves up and at an angle θ_0 (say) with vertical, it stops, this is the extreme position of bob. From work-energy theorem, $\Delta k = W_{ext}$

$$0 - \frac{1}{2}(2m)\left(\frac{v_0}{2}\right)^2 = 2mgl(1-\cos\theta_0)$$

$$\frac{v_0^2}{8gl} = 1-\cos\theta_0 = 2\sin^2\frac{\theta_0}{2}$$

$$\Rightarrow \sin\frac{\theta_0}{2} = \frac{v}{4\sqrt{gl}}$$

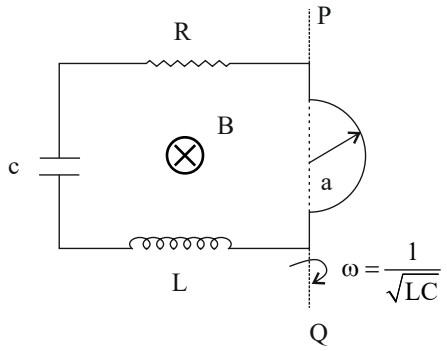
If θ_0 is small, $\frac{\theta_0}{2} = \frac{v_0}{4\sqrt{gl}} \Rightarrow \theta_0 = \frac{v_0}{2\sqrt{gl}}$

So, the equation of simple harmonic motion is

$\theta = \theta_0 \sin\omega t$

$$\theta = \frac{v_0}{2\sqrt{gl}}\sin\left[\sqrt{\frac{g}{l}}t\right]$$

7. A wire shaped as a semicircle of radius a, is rotating about an axis PQ with a constant angular velocity $\omega = \frac{1}{\sqrt{LC}}$, with the help of an external agent. A uniform magnetic field B exists in space and is directed into the plane of the figure. (circuit part remains at rest) (left part is at rest)



(A) The rms value of current in the circuit is $\frac{\pi Ba^2}{R\sqrt{2LC}}$

(B) The rms value of current in the circuit is $\frac{\pi Ba^2}{2R\sqrt{2LC}}$

(C) The maximum energy stored in the capacitor is $\frac{\pi^2B^2a^4}{8R^2C}$

(D) The maximum power delivered by the external agent is $\frac{\pi^2B^2a^4}{4LCR}$

Ans. BC

Sol. Let at time t the angle between magnetic field and area vector (semicircle) be θ, the $\theta = wt$

$$\phi = \vec{B}.\vec{S} = \frac{\pi a^2 B}{2}\cos\omega t$$

$$\varepsilon = -\frac{d\phi}{dt} = \frac{\pi B a^2 \omega}{2}\sin\omega t$$

$$\varepsilon_0 = \frac{\pi B a^2}{2\sqrt{LC}} \text{ peak emf}$$

Since the circuit is in resonance,

$$|z| = R \Rightarrow i_0 = \frac{\pi B a^2}{2R\sqrt{LC}} \text{ peak current}$$

$$i_{rms} = \frac{i_0}{\sqrt{2}} \Rightarrow i_{rms} = \frac{\pi B a^2}{2R\sqrt{2LC}}$$

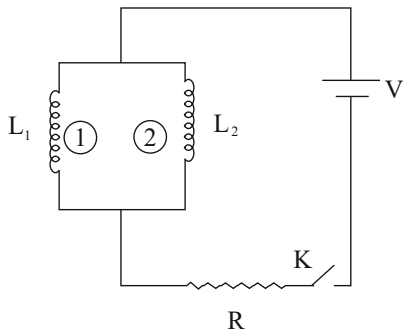
$$U_C = \frac{1}{2}CV_0^2 \rightarrow \text{max energy}, \; V_0 \rightarrow \text{peak voltage}$$

$$V_0 = i_0 X_c = \frac{i_0}{C\omega} = \frac{i_0\sqrt{LC}}{C}$$

$$U_C = \frac{1}{2}C \times \frac{\pi^2 B^2 a^4}{4R^2C^2} = \frac{\pi^2 B^2 a^4}{8R^2C}$$

$$P_{Ext} = P_{Dissipated} = \varepsilon_0 i_0 = \frac{\pi B a^2}{2\sqrt{LC}} \times \frac{\pi B a^2}{2R\sqrt{LC}}, P_{Ext} = \frac{\pi^2 B^2 a^4}{4LCR}$$

8. In the figure shown the key is switched on at t = 0. Let I_1 and I_2 be the currents through inductors having self inductances L_1 & L_2 at any time t respectively. The magnetic energy stored in the inductors 1 and 2 be U_1 and U_2. Then $\frac{U_1}{U_2}$ at any instant of time is :



(A) $\frac{L_1}{L_2}$

(B) $\frac{L_2}{L_1}$

(C) $\frac{I_1}{I_2}$

(D) $\frac{I_2}{I_1}$

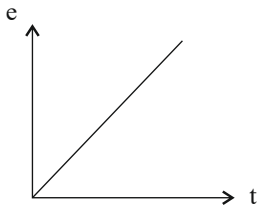
Ans. BC

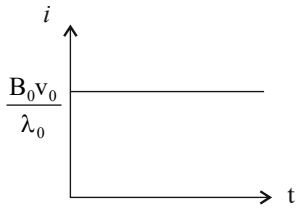
Sol. Because both inductors are in parallel

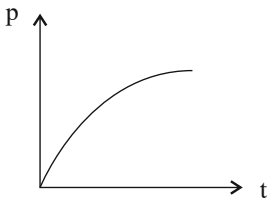
$$\therefore L_1I_1 = L_2I_2$$

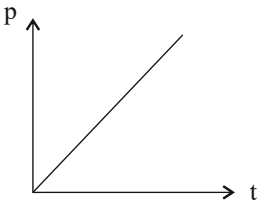
$$\frac{U_1}{U_2} = \frac{1/2L_1I_1I_1}{1/2L_2I_2I_2} = \frac{I_1}{I_2} = \frac{L_2}{L_1}$$

9. A long straight slider is moved with constant velocity $\vec{v} = v_0\hat{i}$ parallel to earth surface in the x-y plane in a uniform magnetic field $\vec{B} = B_0\hat{k}$ on a resistanceless rails in the form of curve $y^2 = 4x$. Here emf induced in the loop, current, power, resistance of slider per unit length and time are represented by e, i, p, λ_0 and t respectively. If at t = 0 slider starts from x = 0. Then which of the following option(s) is / are correct.

(A) 

(B) 

(C) 

(D) 

Ans. BC

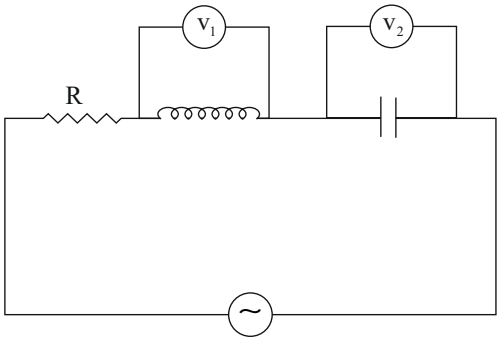
Sol. $e = B_0v_0(2y) = 4B_0v_0^{3/2}\sqrt{t}$

$$i = \frac{e}{\lambda(2y)} = \frac{Bv_0}{\lambda} \qquad \text{where } \Rightarrow y = \sqrt{4v_0t}$$

$$F = \left(\frac{B_0v_0}{\lambda}\right)(2y)B_0 = \frac{2B_0^2v_0y}{\lambda} = \frac{4B_0^2v_0^{3/2}}{\lambda}\sqrt{t}$$

$$p = \vec{F}.\vec{v} = \frac{2B_0^2v_0^2y}{\lambda} = \frac{4B_0^2v_0^{5/2}}{\lambda}\sqrt{t}$$

10. In the figure shows $R=100\Omega, L=\frac{2}{\pi}H$ and $C=\frac{8}{\pi}\mu F$ are connected in series with an a.c source of 200 volt and frequency f. V_1 and V_2 are two hot-wire voltmeters. If the reading of V_1 and V_2 are same, then



(A) $f = 125$ Hz

(B) $f = 250\pi$ Hz

(C) Current through R is 2A

(D) $V_1 = V_2 = 1000$ volt

Ans. ACD

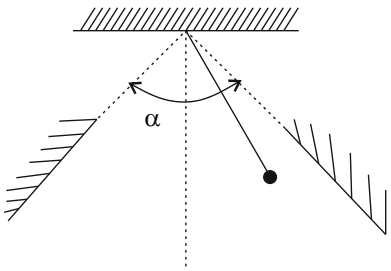
Sol. $V_1 = V_2 \Rightarrow x_L = x_C$

$$\Rightarrow f = \frac{1}{2\pi\sqrt{LC}} = 125\text{Hz}$$

$$I_0 = \frac{v_0}{R} = \frac{200}{100} \qquad (\because x = 0, \therefore z = R)$$

$$V_1 = V_2 = IX_L = I.(\omega L) = I.L.\frac{1}{\sqrt{LC}} = I\sqrt{\frac{L}{C}} = 2\sqrt{\frac{(2/\pi)}{(8\times10^{-6}/\pi)}} = 1000\text{volt}$$

11. A simple pendulum initially oscillating simple harmonically with angular amplitude α and time period T_0 is symmetrically confined between two rigid fixed planes P and Q making angle 'α' with each other as shown in figure. [Assume 'α' is small]



(A) If collision at both walls are elastic then time period is $\frac{T_0}{2}$

(B) If collision at both walls are inelastic then time period is T_0

(C) If collision at one wall is elastic and at other is inelastic, then 1 time period is T_0

(D) If collision at one wall is elastic and at other is inelastic, then 1 time period will be less than T_0.

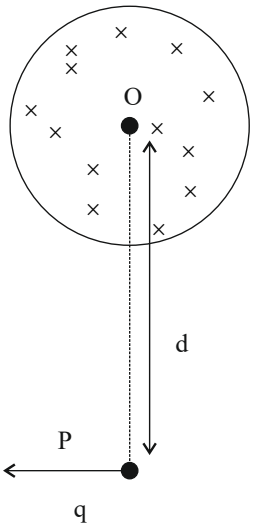
Ans. BC

Sol. Bob will move between $-\frac{\alpha}{2}$ and $\frac{\alpha}{2}$

$$T = 4\left(\frac{T_0}{6}\right) = \frac{2T_0}{3}$$

when collision with at least one wall is inelastic then amplitude will reduce and time period will remain same as T_0

12. In the cylindrical region of radius R = 2m, there exists a time varying magnetic field B such that $\frac{dB}{dt} = 2$ Tesla / sec . A charged particle having charge q = 2C is placed at the point P at a distance d from its center O. Now, the particle is moved in the direction perpendicular to line OP by an external agent up to infinity so that there is no gain in kinetic energy of charged particle. Then



(A) Work done by external agent is 4π Joule when d = 4m

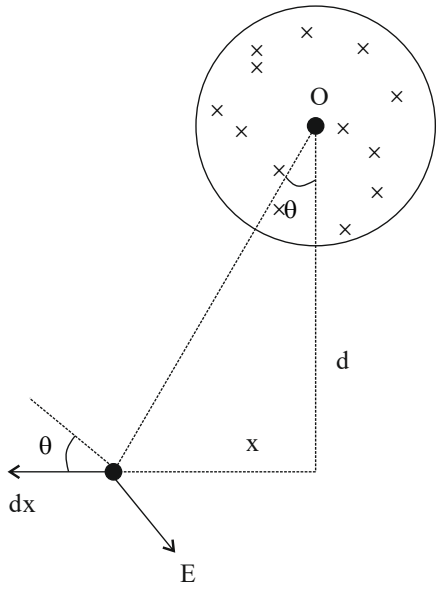
(B) Work done by external agent is 4π Joule when d = 8 m

(C) Work done by external agent is independent of d

(D) Work done by external agent is positive

Ans. ABCD

Sol.



$$\int \vec{E} \cdot d\vec{l} = A.\frac{dB}{dt}$$

$$E.2\pi\sqrt{x^2+d^2} = \pi R^2 k, where\left(\frac{dB}{dt}=k\right)$$

$$E = \frac{R^2 k}{2\sqrt{x^2+d^2}}$$

$$W = \int_0^\infty q\vec{E}.\vec{dx}$$

$$= \int_0^\infty qE\cos\theta.\,dx = \int_0^\infty \frac{qR^2k}{2\sqrt{x^2+d^2}}\frac{d}{\sqrt{d^2+x^2}}dx = \frac{qR^2k\pi}{4}$$

Alternate solution

By applying Faraday's Law for an imaginary triangular loop having vertex O, P and point at infinity, $q\int \vec{E} \cdot \vec{dl}$ can be found out.

13. A uniform conducting ring of mass m = 2 kg, radius r = 2m and resistance 8Ω is kept on smooth, horizontal surface. A time varying magnetic field $\vec{B} = \left(\hat{i} + t^2\hat{j}\right)$ Tesla is present in the region, where t is time in second and take vertical as y-axis. (Take $\pi^2 = 10$). Then

(A) Time when ring starts toppling is 1 sec

(B) Time when ring starts toppling is 3/4 sec

(C) Heat generated through the ring till the instant when the ring start toppling is 40/3 Joule

(D) Heat generated through the ring till the instant when the ring start toppling is 80/3 Joule

Ans. AD

Sol. $\phi = BA = \left(t^2\right)\left(\pi r^2\right)$

$$\varepsilon = \frac{d\phi}{dt} = 2\pi r^2 t$$

$$i = \frac{\varepsilon}{R} = \frac{2\pi r^2 t}{R}$$

For toppling

$$\tau = \left|\vec{\mu}\times\vec{B}\right| \geq mgr$$

$$\left(i.\,\pi r^2\right).1 \geq mgr$$

$$\frac{2\pi r^2 t}{R}.\pi r^2 \geq mgr$$

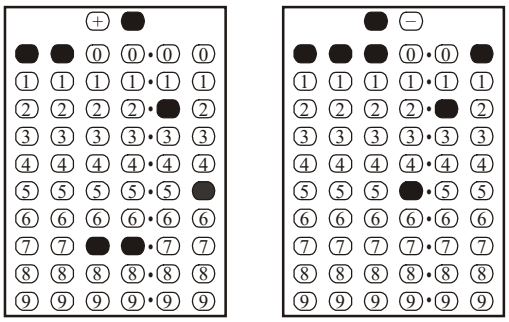
$$t \geq \frac{mgR}{2\pi^2 r^3} = 1\text{sec}$$

$$\text{Heat} = \int i^2 R dt = \int\left(\frac{2\pi r^2 t}{R}\right)^2 dt.\,R$$

$$\text{Heat} = \frac{4\pi^2 r^4}{R}\int_0^1 t^2 dt = \frac{80}{3} Joule$$

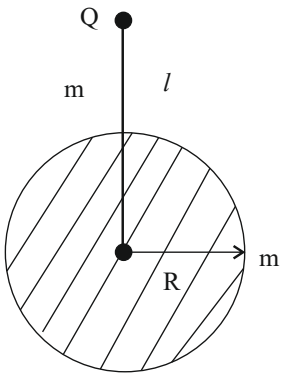
SECTION-II : (Maximum Marks: 20)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.
  For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +4 If ONLY the correct numerical value is entered as answer.
  Zero Marks : 0 In all other cases.

14. A meter stick connected with disc of same mass which is joined at one end of stick and disc can freely rotates about its joint. System swings in a vertical plane about a fixed horizontal axis passing through its one end, undergoes small oscillation of frequency ' f_0 '. If disc is removed from the stick, then its new frequency of small oscillation is $\sqrt{\frac{35}{k}}f_0$, find the value of k. (take R = l/2)



Ans. 24

Sol. We know that, time period of a compound pendulum is given by

$$T = 2\pi\sqrt{\frac{I}{mgd}} \Rightarrow f = \frac{1}{2\pi}\sqrt{\frac{mgd}{I}}$$

where, I = moment of inertia about hinge

$$I = I_{rod,Q} + I_{Disc,Q}$$

$$= \frac{ml^2}{3} + \frac{mR^2}{2} + ml^2 = \frac{ml^2}{3} + \frac{ml^2}{8} + ml^2$$

$$I = \frac{35}{24}ml^2$$

and $d = \frac{3l}{4}$ {Distance of CM of the system from hinge}

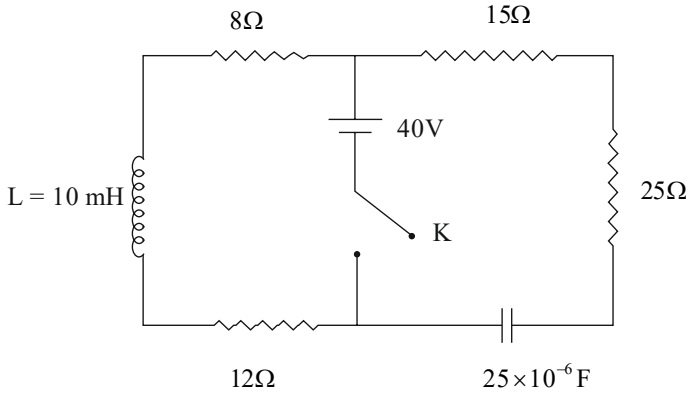
$$\therefore f_0 = \frac{1}{2\pi}\sqrt{\frac{(2m)g(3l/4)}{\left(\frac{35}{24}ml^2\right)}} = \frac{1}{2\pi}\sqrt{\frac{36}{35}\frac{g}{l}}$$

when disc is removed

$$f = \frac{1}{2\pi}\sqrt{\frac{mg\left(\frac{l}{2}\right)}{\frac{ml^2}{3}}} = \frac{1}{2\pi}\sqrt{\frac{3g}{2l}}$$

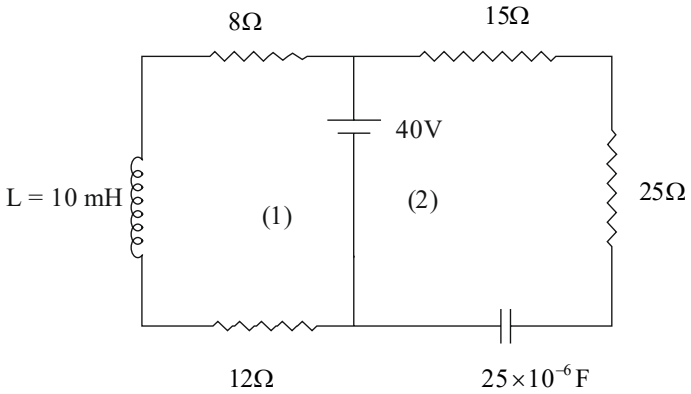
$$\frac{f}{f_0} = \sqrt{\frac{3}{2}\frac{35}{36}} \Rightarrow f = \sqrt{\frac{35}{24}}f_0$$

15. In the circuit the key (K) is closed at t = 0. Find the current in ampere through the key (K) at the instant $t = 10^{-3}\ln 2\,\text{sec}$



Ans. 2

Sol.



i_1 is current in the first loop

$$i_1 = \frac{40}{20}\left(1 - e^{-\frac{t}{5\times10^{-4}}}\right) = \frac{3}{2}amp$$

i_2 is current in the second loop

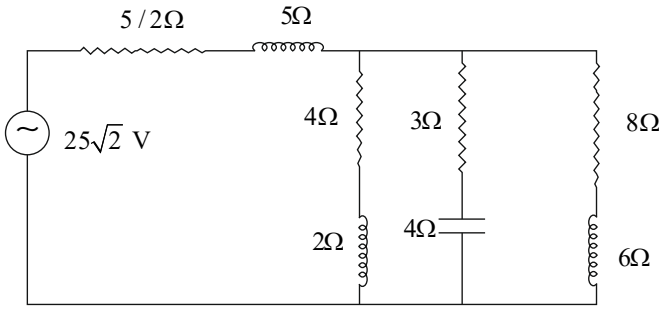
$$i_2 = \frac{40}{40} e^{-\frac{t}{10^{-3}}} = \frac{1}{2} amp$$

So, total current through key (K)

$$i = i_1 + i_2$$

$$= \frac{3}{2} + \frac{1}{2} = 2amp$$

16. Consider a network shown in figure. The current supplied by the AC voltage source is n amperes. What is the value of n.



Ans. 5

Sol. $\frac{1}{Z_{eq}} = \frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3}$ (for parallel circuit)

$$= \frac{1}{4+2i} + \frac{1}{3-4i} + \frac{1}{8+6i} = \frac{4-2i}{20} + \frac{3+4i}{25} + \frac{8-6i}{100}$$

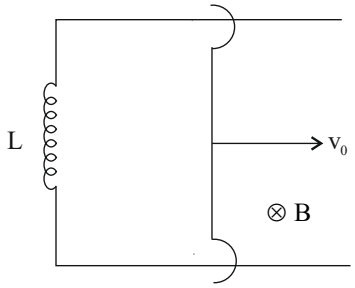
Solving this we will get $z_{eq} = \frac{5}{2} + 0i$

Now, over all impedance of the circuit is $z = \frac{5}{2} + \frac{5}{2} + 5i = 5 + 5i$

Therefore, $|z| = 5\sqrt{2}$

Current supplied by source $= \frac{25\sqrt{2}}{5\sqrt{2}} = 5$

17. A conducting rod of mass m = 0.3 kg and length $l = 4\ m$ can slide without friction on two parallel conducting rails. The conducting rails are connected via an inductance L = 3mH. This system is placed in a region containing uniform magnetic field B = 1 T pointing into the plane. If the rod is given an initial velocity $v_0 = 2\ m/s$, its oscillates with an amplitude A cm. Find the value of 2A.



Ans. 3

Sol. $L\frac{di}{dt} = Blv$

$$m\frac{dv}{dt} = -iBl$$

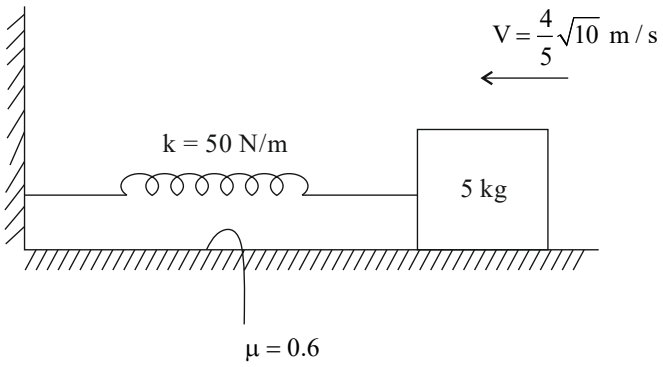
$$\Rightarrow \frac{d^2v}{dt^2} = -\left(\frac{B^2l^2}{mL}\right)v \Rightarrow v = v_0\cos(\omega t)$$

with $\omega = \frac{Bl}{\sqrt{mL}}$

Thus, $\frac{dx}{dt} = v_0\cos(\omega t) \Rightarrow x = \frac{v_0}{\omega}\sin\omega t$

$$2A = \frac{2v_0}{\omega} = 3cm$$

18. A spring block system is placed on a horizontal rough surface as shown in the figure. The block is given velocity when the spring is in natural length. The total distance travelled by the block before it finally comes to rest is



Ans. 0.4

Sol. Applying energy conservation when spring is maximum compressed

$$-\mu mgx - \frac{1}{2}kx^2 = 0 - \frac{1}{2}mv^2$$

$$\Rightarrow -\frac{3}{5}(50)x - \frac{1}{2}(50)x^2 = -\frac{1}{2}\times 5\times\left(\frac{4}{5}\sqrt{10}\right)^2$$

$$\Rightarrow 30x + 25x^2 = 16$$

or $25x^2 + 30x - 16 = 0 \Rightarrow x = \frac{2}{5}meters$

So at the maximum compression spring force $= 50\left(\frac{2}{5}\right) = 20\text{N}$

and friction force $= \frac{3}{5}\times(5\times 10) = 30\text{N}$

Hence the block will not move so distance $= \frac{2}{5}\text{m}$

PART-2 : CHEMISTRY

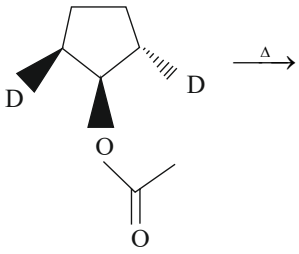
PART-2 : CHEMISTRY

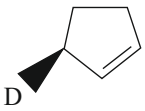
SECTION–I(i) : (Maximum Marks : 15)

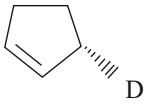
- This section contains **FIVE** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

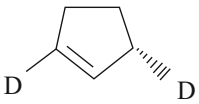
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases

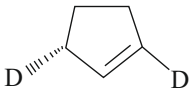
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1.  major product is :

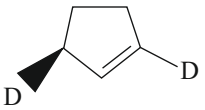
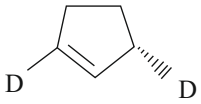
(A) 

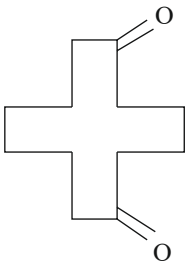
(B) 

(C) 

(D) 

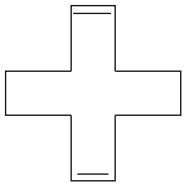
Ans. C

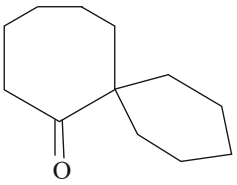
Sol. Pyrolysis of eter is syn elimination and $C-D$ bond is stronger than $C-H$ so  is formed which can also be written as 

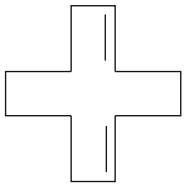
2. 

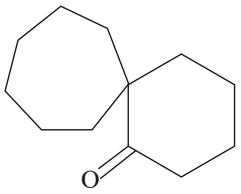
$$\xrightarrow[(ii)\,H_2O]{(i)\,Mg-Hg} A \xrightarrow[\Delta]{conc.H_2SO_4} B\,(Major)$$

'B' is :

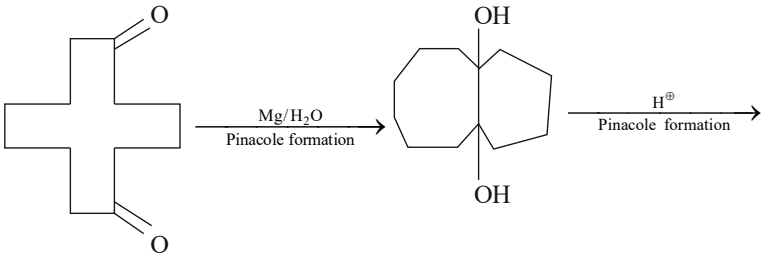
(A) 

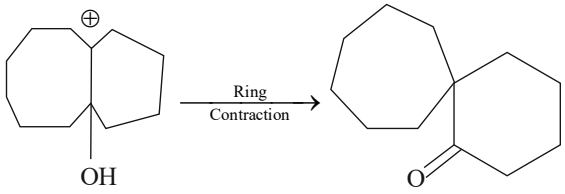
(B) 

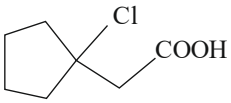
(C) 

(D) 

Ans. D

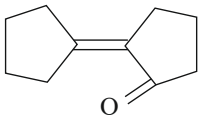
Sol. 

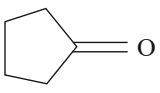


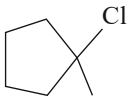
3. 

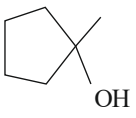
$$\xrightarrow[(ii)\,dil\,OH^{\ominus},\Delta]{(i)\,NaOH,CaO,\Delta\ (ii)\,H^{\oplus}/KMnO_4,\Delta} 'X'$$

'X' obtained :

(A) 

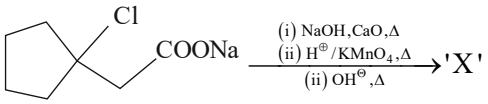
(B) 

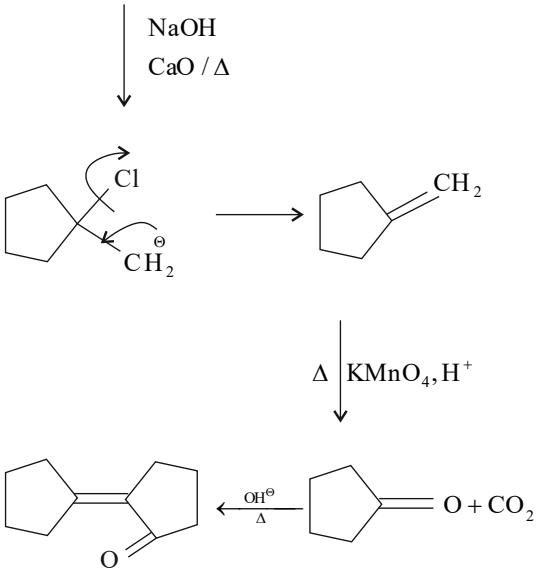
(C) 

(D) 

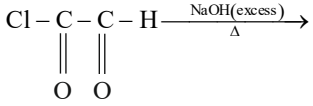
Ans. A

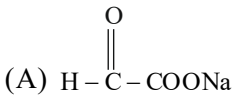
Sol.

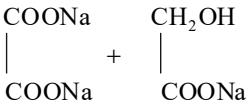


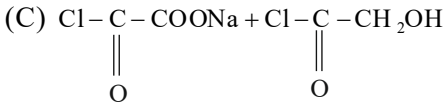


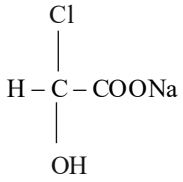
4.



(A) 

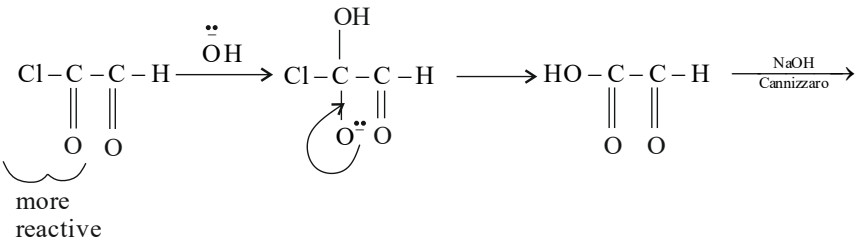
(B) 

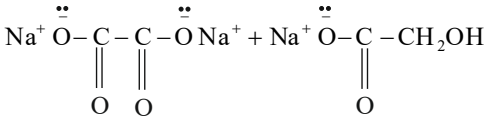
(C) 

(D) 

Ans. B

Sol.





5. When 1 mol of a mono atomic ideal gas at T K undergoes adiabatic change under a constant external pressure of 1 atm, changes volume from 1 L to 2 L. The final temperature (in K) would be

(A) $\frac{T}{2^{2/3}}$ (B) $T+\frac{2}{3\times 0.0821}$

(C) T (D) $T-\frac{2}{3\times 0.0821}$

Ans. D

Sol. Process $\Delta H = 0$

$\Delta E = \Delta W$

$\Delta W = -P\Delta V = -1$ lit atm

$\Delta E = nC_V\Delta T$

$C_V = \frac{R}{\gamma - 1}, \gamma = \frac{5}{3}$ for mono atomic gas

$\left(\gamma = \frac{C_P}{C_V}\right)$

$C_V = \frac{R}{\frac{5}{3}-1} = \frac{3R}{2}$

$n = 1$

$\Delta E = 1\times\frac{3R}{2}\times(T_2 - T) = -1$

$T_2 = T - \frac{2}{3\times 0.0821}$

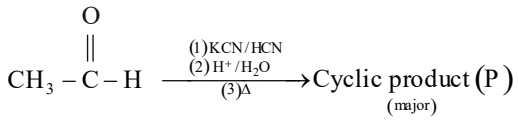
SECTION–I(ii) : (Maximum Marks : 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

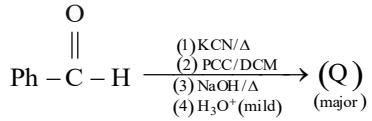
Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

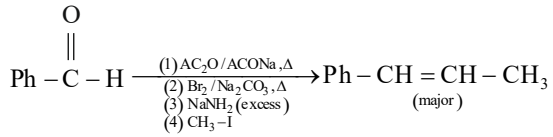
6. Which of the following is / are correctly matched

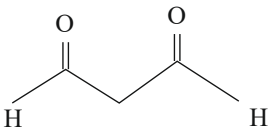
(A) 

No.of stereomers of (P) = 3

(B) 

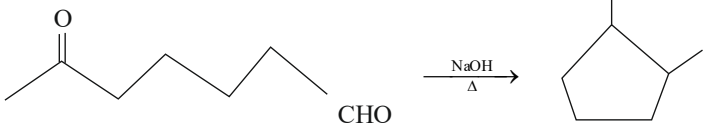
Then, 1 mole of (Q) liberates 44.8 lit of CH_4 gas with excess of CH_3MgBr / ether , at N.T.P

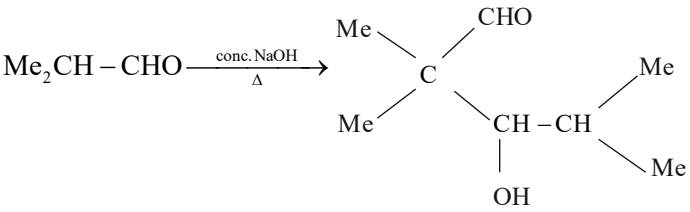
(C) 

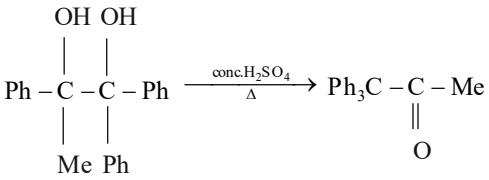
(D)  gives yellow precipitation with $NaOBr_{(aq)}$

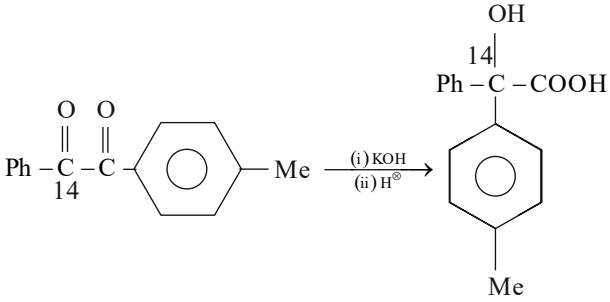
Ans. AB

7. Select option(s) in which correct major product is not given :

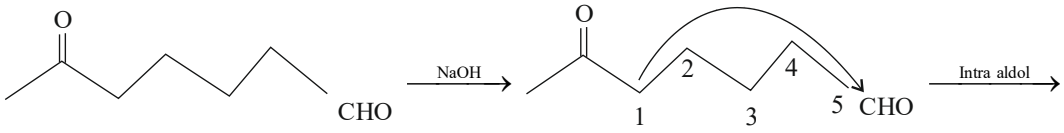
(A) 

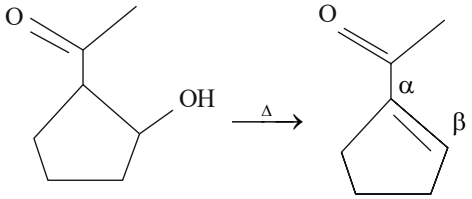
(B) 

(C) 

(D) 

Ans. ABD

Sol. A. 

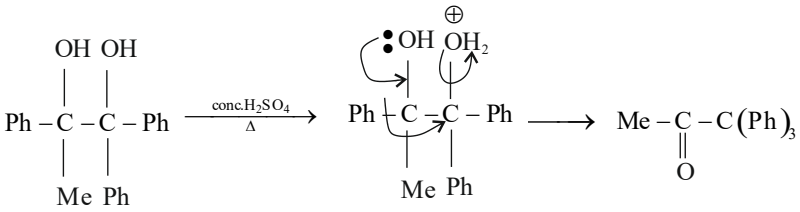


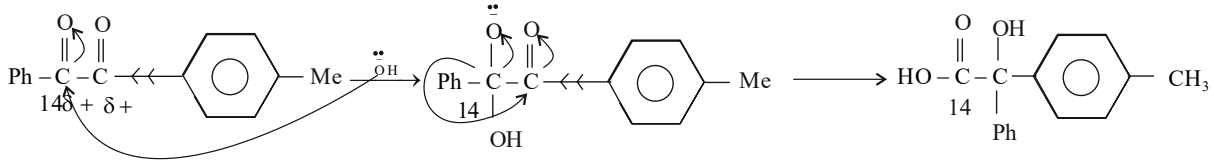
(Major)

$\therefore$ Reactivity : aldehyde > ketone

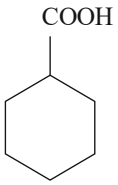
B. $Me_2CH-CHO \xrightarrow[\Delta]{conc.KOH}$ Canniozzaro reaction

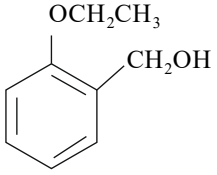
Due to hindrane on carbonyl, aldol does not happen. We get $Me_2CH-CH_2OH + Me_2CH-\underset{\underset{O}{\|}}{C}-ONa$

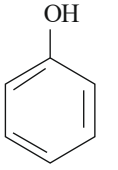
C. 

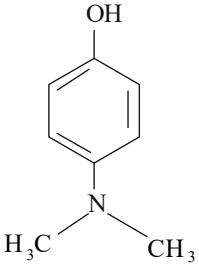
D. 

8. Amongst the following compound(s) which is / are soluble in aqueous NaOH ?

(A) 

(B) 

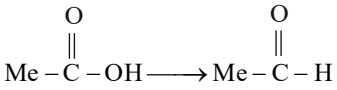
(C) 

(D) 

Ans. ACD

Sol. Conceptual

9. For the given conversion :



Suitable options are :

(A) Reduction by $LiAlH_4$

(B) Reaction with $SOCl_2$ followed by $H_2 / Pd - BaSO_4$

(C) Catalytic reduction by H_2 / Ni

(D) Reaction with PCl_5 followed by $DIBAL - H / -78°C$

Ans. BD

Sol. Conceptual

10. Compound(s) without carboxylic acid group in it are :

(A) Barbituric acid (B) Carbolic acid

(C) Picric acid (D) Cyanic acid

Ans. ABCD

Sol. Conceptual

11. The normal boiling point of a liquid X is 400 K. ΔH_{vap} at normal boiling point is 40 kJ/mol. Select correct statement(s) :

(A) $\Delta S_{vaporisation} < 100\,J/mol.K$ at 400 K and 2 atm

(B) $\Delta S_{vaporisation} < 10\,J/mol.K$ at 400 K and 1 atm

(C) $\Delta G_{vaporisation} < 0$ at 410 K and 1 atm

(D) $\Delta U = 43.32\ kJ/mol.K$ at 400 K and 1 atm

Ans. AC

Sol. $\Delta S_{vap} = \dfrac{\Delta H}{B.P} = \dfrac{40\times1000}{400} = 100\ J/mole$

(A) $P = 2$ atm $\quad B.P > 400\ K$

$P \uparrow\ B.P \uparrow \quad \Delta S_{vap} < 100$

(C) $\Delta G = \Delta H - T\Delta S$

$= 40\times1000 - 410\times100 = -ve$

$\Delta G < 0$

(D) $\Delta H = \Delta U + \Delta n_g RT$

$40 = \Delta U + 1\times8.31\times10^{-3}\times400$

$\Delta U = 36.676\ kJ/mol$

12. For an isolated system, the entropy :

(A) Either increases or remains constant (B) Either decreases or remains constant

(C) Can never decrease (D) Can never increase

Ans. AC

Sol. Entropy of isolated system increases or remains constant but it can never decreases.

13. $Me-\underset{\underset{O}{\|}}{C}-Et + Me-\underset{\underset{O}{\|}}{C}-Ph \xrightarrow[(ii)\,H_2O]{(i)\,Mg(Hg)} Product\,(s)$

Product (s) is / are :

(A) $Me-\underset{\underset{OH}{|}}{\overset{\overset{Et}{|}}{C}}-\underset{\underset{OH}{|}}{\overset{\overset{Ph}{|}}{C}}-Me$

(B) $Me-\underset{\underset{OH}{|}}{\overset{\overset{Me}{|}}{C}}-\underset{\underset{OH}{|}}{\overset{\overset{Ph}{|}}{C}}-Et$

(C) $Me-\underset{\underset{OH}{|}}{\overset{\overset{Et}{|}}{C}}-\underset{\underset{OH}{|}}{\overset{\overset{Et}{|}}{C}}-Me$

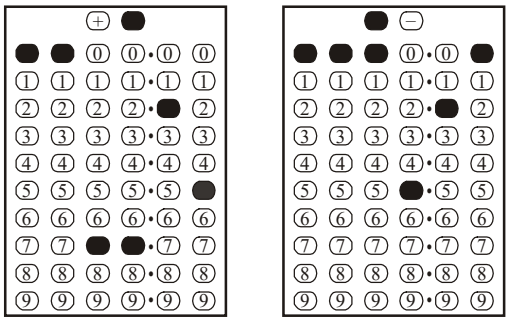
(D) $Me-\underset{\underset{OH}{|}}{\overset{\overset{Me}{|}}{C}}-\underset{\underset{OH}{|}}{\overset{\overset{Ph}{|}}{C}}-Ph$

Ans. ACD

Sol. $Me-\underset{\underset{O}{\|}}{C}-Et + Me-\underset{\underset{O}{\|}}{C}-Ph \xrightarrow[(ii)H_2O]{(i)Mg(Hg)} Me-\underset{\underset{OH}{|}}{\overset{\overset{Et}{|}}{C}}-\underset{\underset{OH}{|}}{\overset{\overset{Ph}{|}}{C}}-Me + Me-\underset{\underset{OH}{|}}{\overset{\overset{Et}{|}}{C}}-\underset{\underset{OH}{|}}{\overset{\overset{Et}{|}}{C}}-Me + Me-\underset{\underset{OH}{|}}{\overset{\overset{Ph}{|}}{C}}-\underset{\underset{OH}{|}}{\overset{\overset{Me}{|}}{C}}-Ph$

SECTION-II : (Maximum Marks: 20)

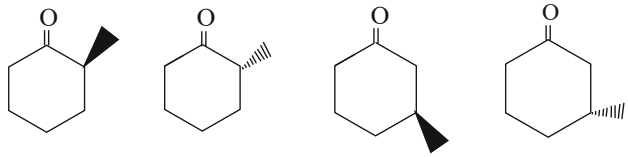
- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.
  For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



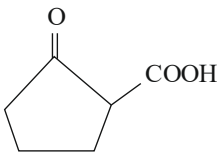
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +4 If ONLY the correct numerical value is entered as answer.
  Zero Marks : 0 In all other cases.

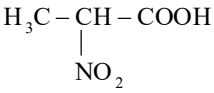
14. The number of compounds (molar mass = 112) with six membered ring, show '+' ve Brady's test and give optically active product with $NaBH_4$ = 'X' (Including stereomers) then the value of '12X' is

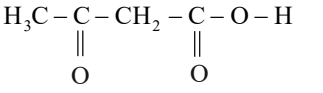
Ans. 48

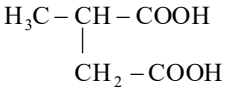
Sol. 

15. Number of compounds, which can evolve CO_2 on heating, are Z, what is $Z\times 5=.....$









$HOOC-CH_2-COOH$

Ans. 20

16. An unknown compound having molecular formula $C_5H_{10}O_5$ was reacted with excess of CH_3COCl and pyridine to give compound having molecular weight 318 gm. The number of hydroxy groups present in the unknown compound is Z , then the value of $(Z\times 15)$ is

Ans. 60

Sol. $C_5H_{10}O_5 \overset{M.wt}{\Rightarrow} 150g$

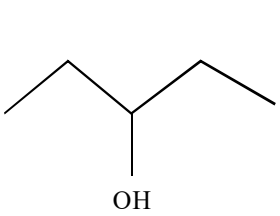
After reaction with CH_3COCl its molecular weight is 318 g reaction with one CH_3COCl molecule increases M.wt of the compound by 42g

Overall increase in M.wt of the comp is 168 g

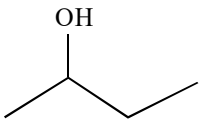
$$Z=\frac{168}{42}=4$$

$4\times 15=60$

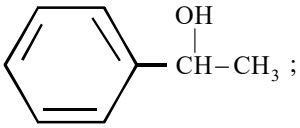
17. 'x' number of alcohols among the following will give positive Iodoform test. Then $x\times 10=?$



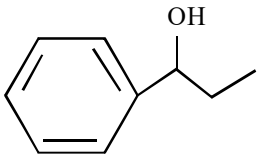
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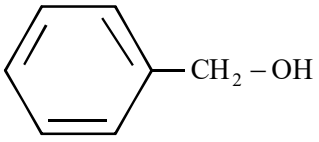
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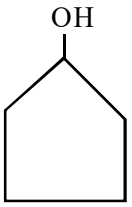
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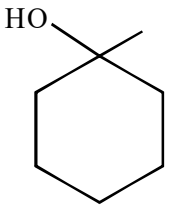
CH_3-CH_2-OH ;



;

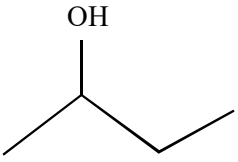


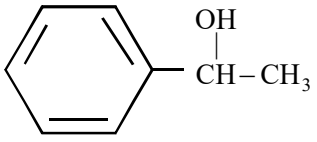
;



Ans. 30 ; x = 3

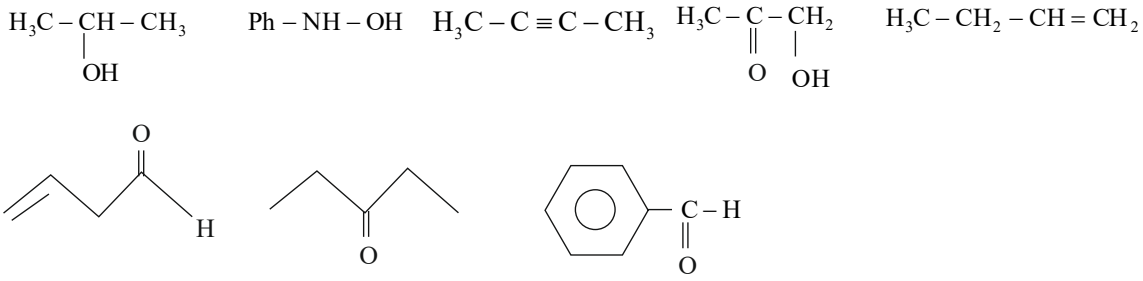
Sol.





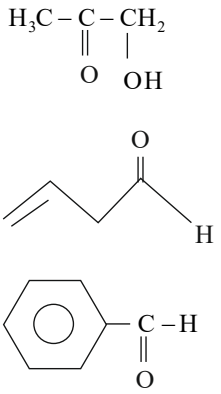
CH_3-CH_2-OH

18. Number of compounds from below which can reduce Tollen's reagent is x. What is $(x)\times 11$?



Ans. 44

Sol. $Ph-NH-OH$



PART-3 : MATHEMATICS

SECTION–I(i) : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –1 In all other cases

---

1. If x is real, then the maximum value of $y = 2(a-x)\left(x+\sqrt{x^2+b^2}\right)$ is $(\text{given } a \neq 0)$

(A) a^2+b^2 (B) a^2

(C) $\sqrt{a^2+b^2}$ (D) $a\sqrt{a^2+b^2}$

Ans. A

Sol. Let $t = x+\sqrt{x^2+b^2}$

$$\Rightarrow \frac{1}{t} = \frac{1}{x+\sqrt{x^2+b^2}} = \frac{\sqrt{x^2+b^2}-x}{b^2}$$

$$\therefore t - \frac{b^2}{t} = 2x \text{ and } t+\frac{b^2}{t} = 2\sqrt{x^2+b^2}$$

Then $2(a-x)\left(x+\sqrt{x^2+b^2}\right) = \left(2a-t+\frac{b^2}{t}\right).(t)$

$$= 2at - t^2 + b^2 = a^2+b^2-\left(a^2-2at+t^2\right)$$

$$= a^2+b^2-(a-t)^2$$

Therefore, $y = 2(a-x)\left(x+\sqrt{x^2+b^2}\right) \le a^2+b^2$. Hence, the maximum value of y is a^2+b^2. This value is attained at $t = a$ or $x = \left(a^2-b^2\right)/2a$

2. If tanA and tan B are the roots of the quadratic equation $x^2-px+q=0$, then value of $\sin^2(A+B)$ is

(A) $\frac{p^2}{p^2+q^2}$ (B) $\frac{p^2}{(q+p)^2}$

(C) $1-\frac{p}{(1-q)^2}$ (D) $\frac{p^2}{(1-q)^2+p^2}$

Ans. D

Sol. We have tan A + tan B = p, and tanA tanB=q.

$$\tan(A+B) = \frac{\tan A+\tan B}{1-\tan A\tan B} = \frac{p}{1-q}$$

Now, $\sin^2(A+B)=\dfrac{\tan^2(A+B)}{\sec^2(A+B)}$

$$=\frac{p^2/(1-q)^2}{1+p^2/(1-q)^2}=\frac{p^2}{(1-q)^2+p^2}$$

3. For a, b > 0, the value of $\int_0^1 \dfrac{\log\left(\dfrac{(x+a)^{(x+a)}}{(x+b)^{(x+b)}}\right)}{(x+a)(x+b)\log(x+a)\log(x+b)}dx$ is equal to

(A) $\log\left(\dfrac{1+a}{1+b}\right)\dfrac{\log a}{\log b}$

(B) $\dfrac{\log(1+a)\log a}{\log(1+b)\log b}$

(C) $\log\left(\dfrac{\log(1+a)\log a}{\log(1+b)\log b}\right)$

(D) $\log\left(\dfrac{\log(1+b)\log a}{\log(1+a)\log b}\right)$

Ans. D

Sol. $\dfrac{\log\left(\dfrac{(x+a)^{(x+a)}}{(x+b)^{(x+b)}}\right)}{(x+a)(x+b)\log(x+a)\log(x+b)}$

$$=\frac{(x+a)\log(x+a)-(x+b)\log(x+b)}{(x+a)(x+b)\log(x+a)\log(x+b)}$$

$$=\frac{1}{(x+b)\log(x+b)}-\frac{1}{(x+a)\log(x+a)}$$

Required Integral

$$=\int_0^1\frac{1}{(x+b)\log(x+b)}dx-\int_0^1\frac{dx}{(x+a)\log(x+a)}$$

$$=\log\left|\log(x+b)_0^1\right|-\log\left|\log(x+a)\right|_0^1$$

$$=\log\frac{\log(1+b)}{\log b}-\log\frac{\log(1+a)}{\log a}$$

$$=\log\left(\frac{\log(1+b)\log a}{\log(1+a)\log b}\right)$$

4. Through any point (x, y) of a curve which passes through the origin, lines are drawn parallel to the coordinate axes. The curve, given that it divides the rectangle formed by the two lines and the axes into two areas, one of which is twice the other, represents a family of

(A) Circles

(B) Parabolas

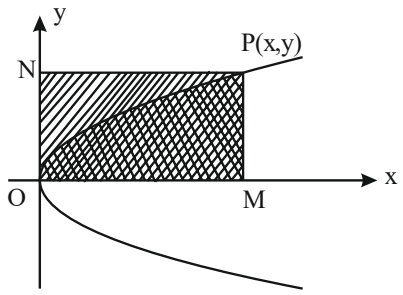
(C) Hyperbolas

(D) Straight lines

Ans. B

Sol. Let P(x, y) be the point on the curve passing through the origin O(0, 0), and let PN and PM be the lines parallel to the x-and y - axes, respectively (figure). If the equation of the curve is y = y(x), the area POM equals $\int_0^x y\,dx$ and the area PON equals $xy-\int_0^x y\,dx$. Assuming that $2(\text{POM})=\text{PON}$, we therefore have

$$2\int_0^x y\,dx = xy-\int_0^x y\,dx \Rightarrow 3\int_0^x y\,dx = xy$$



Differentiating both sides of this gives

$$3y = x\frac{dy}{dx}+y \Rightarrow 2y = x\frac{dy}{dx} \Rightarrow \frac{dy}{y}=2\frac{dx}{x}$$

$\Rightarrow \log|y| = 2\log|x|+C \Rightarrow y = Cx^2$, with C being a constant.

This solution represents a parabola. We will get a similar result if we had started instead with $2(\text{PON})=\text{POM}$.

5. Consider the following regions in the plane

$$R_1=\{(x,y): 0\le x\le 1 \text{ and } 0\le y\le 1\}$$

$$R_2=\{(x,y): x^2+y^2\le 4/3\}$$

The area of the region $R_1\cap R_2$ can be expressed as $\frac{a\sqrt{3}+b\pi}{9}$, where a and b are integers. Then the value of (a + b) equals

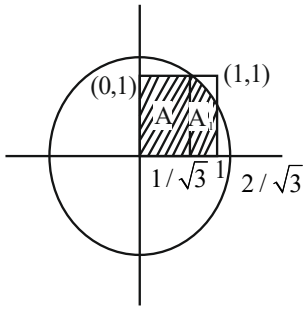
(A) 2 (B) 3

(C) 4 (D) 5

Ans. C

Sol. $A=\frac{1}{\sqrt{3}}\times 1+A_1$

$$A_1=\int_{1/\sqrt{3}}^{1}\left(\sqrt{(4/3)-x^2}\right)dx$$



Put $x = \frac{2}{\sqrt{3}}\sin\theta \Rightarrow dx = \frac{2}{\sqrt{3}}\cos\theta d\theta$

$$= \int_{\pi/6}^{\pi/3} \frac{2}{\sqrt{3}}\cos\theta.\frac{2}{\sqrt{3}}\cos\theta\, d\theta = \frac{2}{3}\int_{\pi/6}^{\pi/3}(\cos 2\theta + 1)d\theta$$

$$= \frac{2}{3}\left[\frac{1}{2}\sin 2\theta + \theta\right]_{\pi/6}^{\pi/3}$$

$$= \frac{2}{3}\left[\frac{1}{2}.\frac{\sqrt{3}}{2} + \frac{\pi}{3} - \frac{1}{2}.\frac{\sqrt{3}}{2} - \frac{\pi}{6}\right] = \frac{2}{3}.\frac{\pi}{6} = \frac{\pi}{9}$$

$$A = \frac{1}{\sqrt{3}} + \frac{\pi}{9} = \frac{3\sqrt{3}+\pi}{9}$$

$a = 3, b = 1$

$a + b = 4$

SECTION - I (ii) : (Maximum Marks : 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

6. Let a, b, c, p, q be five different non-zero real numbers and x, y, z be three numbers satisfying the system of equations

$$\frac{x}{a}+\frac{y}{a-p}+\frac{z}{a-q}=1$$

$$\frac{x}{b}+\frac{y}{b-p}+\frac{z}{b-q}=1$$

$$\frac{x}{c}+\frac{y}{c-p}+\frac{z}{c-q}=1$$

then

(A) $x+y+z=a+b+c-p-q$

(B) $x=\frac{abc}{pq}$

(C) $y=\frac{(a-p)(b-p)(c-p)}{p(p-q)}$

(D) $z=\frac{(a-q)(b-q)(c-q)}{q(q-p)}$

Ans. A, B, C, D

Sol. Note that a, b, c are roots of

$$\frac{x}{t}+\frac{y}{t-p}+\frac{z}{t-q}=1$$

$$\Leftrightarrow x(t-p)(t-q)+yt(t-q)+zt(t-p)=t(t-p)(t-q)$$

$$\Leftrightarrow t^3-(x+y+z+p+q)t^2+\left[(p+q)x+qy+pz\right]t-pqx=0$$

From here the answers can be given easily

7. Suppose $m\in R$. The quadratic equation $x^2-(m-3)x+m=0$ has

(A) Real and distinct roots if and only if $m\in(-\infty,1)\cup(9,\infty)$

(B) Both positive roots if and only if $m\in[9,\infty)$

(C) Both negative roots if and only if $m\in(0,1]$

(D) At least one positive root if and only if $m\in(9,\infty)$

Ans. A, B, C

Sol. $x^2-(m-3)x+m=0$ ....(1)

will have real and distinct roots

$$\Leftrightarrow D=(m-3)^2-4m>0$$

$$\Leftrightarrow m^2-10m+9>0$$

$$\Leftrightarrow m\in(-\infty,1)\cup(9,\infty)$$

Roots of (1) will be both positive if

$D\geq 0,\ m-3>0,\ m>0$

$\Rightarrow m\geq 9, m>3$

$\therefore m\in[9,\infty)$

Roots of (1) will be negative if

$D \geq 0, m-3<0, m>0$

$\Rightarrow m \leq 1, 0<m<3$

Thus, $m \in (0,1]$

For $m < 0$, equation has one positive and one negative root.

8. For $k \in N$, let $b_k = \int_1^e \frac{1}{(k+\log x)(k+1+\log x)} \frac{dx}{x}$ and $a_n = \sum_{k=1}^{n} b_k$. Then

(A) $b_k = \log\left(\frac{(k+1)^2}{k(k+2)}\right)$ (B) $a_n = \log\left(\frac{2n+2}{n+2}\right)$

(C) $\lim_{n\to\infty} a_n = \log 2$ (D) $\lim_{n\to\infty} a_n < 1$

Ans. A, B, C, D

Sol. Put $k + \log x = t$, then

$$b_k = \int_k^{k+1} \frac{1}{t(t+1)} dt = \int_k^{k+1} \left[\frac{1}{t} - \frac{1}{t+1}\right] dt$$

$$= \log\left(\frac{k+1}{k}\right) - \log\frac{k+2}{k+1} = \log\frac{(k+1)^2}{k(k+2)}$$

$$a_n = \sum_{k=1}^{n} b_k = \sum_{k=1}^{n} \left[\log\frac{k+1}{k} - \log\frac{k+2}{k+1}\right]$$

$$= \log 2 - \log\frac{n+2}{n+1} = \log\frac{2n+2}{n+2} = \log\left(\frac{2+2/n}{1+2/n}\right)$$

$$\lim_{n\to\infty} a_n = \log 2 < 1$$

9. For $a > 0$, let $I(a) = \int_0^a \frac{dx}{1+e^{a^2-2ax}}$ and let $0 < \theta < \frac{\pi}{2}$, then

(A) $I'(a) = \ln\sqrt{e}$ (B) $I(\sin\theta) + I(\cos\theta) \leq \sin(\pi/4)$

(C) $I(\sin^2\theta) + I(\cos^2\theta) = \frac{1}{2}$ (D) $I(2a - a^2) \leq \frac{1}{2}$

Ans. A, B, C, D

Sol. $I(a) = \int_0^a \frac{dx}{1+e^{a^2-2ax}}$

$$= \int_0^a \frac{e^{x^2}}{e^{x^2} + e^{(a-x)^2}} dx \qquad ...(1)$$

Using $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$, we get

$$I(a) = \int_0^a \frac{e^{(a-x)^2}}{e^{(a-x)^2} + e^{x^2}} dx \qquad ....(2)$$

Adding (1) and (2), we get

$$2I(a)=\int_0^a \frac{e^{x^2}+e^{(a-x)^2}}{e^{x^2}+e^{(a-x)^2}}\,dx=a$$

$$\Rightarrow I(a)=\frac{1}{2}a$$

$$\Rightarrow I'(a)=\frac{1}{2}=\log\sqrt{e}$$

Also, $I(\sin\theta)+I(\cos\theta)$

$$=\frac{1}{2}(\sin\theta+\cos\theta)$$

$$=\frac{1}{\sqrt{2}}\sin\left(\theta+\frac{\pi}{4}\right)\le\frac{1}{\sqrt{2}}$$

Now, $I(\sin^2\theta)+I(\cos^2\theta)=\frac{1}{2}(\sin^2\theta+\cos^2\theta)=\frac{1}{2}$

Next, $I(2a-a^2)=\frac{1}{2}\left[1-(1-a)^2\right]\le\frac{1}{2}$

10. If $y=f(x)$ is a solution of $x\frac{dy}{dx}-y=x\tan\frac{y}{x}$, $y(1)=\frac{\pi}{2}$, then

(A) $\sin\frac{y}{x}=x$ (B) $\int_0^1 y\,dx=\frac{3}{8}\pi$

(C) $\operatorname{cosec}\frac{y}{x}=|x|$ (D) $\int_0^1 y\,dx=\frac{\pi}{8}$

Ans. A, B

Sol. Putting $y=ux$ so that $\frac{dy}{dx}=u+x\frac{du}{dx}$, the given differential equation reduces to $u+x\frac{du}{dx}-u=\tan u$

$$\Rightarrow \cot u\,du=\frac{dx}{x}$$

$$\Rightarrow \log|\sin u|=\log|x|+\log C$$

$$\Rightarrow \left|\sin\frac{y}{x}\right|=C|x|$$

Putting $x=1$, $\left|\sin\frac{\pi}{2}\right|=C\Rightarrow C=1$

So, $\left|\sin\frac{y}{x}\right|=|x|$

$$\sin\frac{y}{x}=\pm x$$

$0 < x < 1, \sin\frac{y}{x} = x \Rightarrow y = x\sin^{-1}x$

$\int_0^1 y\,dx = \int_0^1 x\sin^{-1}x\,dx$

$= \frac{x^2}{2}\sin^{-1}x\Big|_0^1 + \frac{1}{2}\int_0^1 \frac{1-x^2-1}{\sqrt{1-x^2}}dx$

$= \frac{1}{2}\frac{\pi}{2} + \frac{1}{2}\int_0^1 \sqrt{1-x^2}\,dx - \frac{1}{2}\int_0^1 \frac{dx}{\sqrt{1-x^2}}$

$= \frac{1}{2}\frac{\pi}{2} + \frac{1}{2}\sin^{-1}x\Big|_0^1 - \frac{1}{2}\left[\frac{\pi}{2}\sqrt{1-x^2} + \frac{1}{2}\sin^{-1}x\right]_0^1$

$\Rightarrow \frac{\pi}{2} - \frac{1}{4}\frac{\pi}{2} = \frac{3}{8}\pi$

11. Let f be a nonnegative function defined on the interval $[0,\pi/2]$. If

$\int_0^x (f'(t) - \sin 2t)dt = \int_x^0 f(t)\tan t\,dt$ and $f(0) = 1$, then

(A) f(0) = 1 is the absolute maximum value of f (B) f(0) = 1 is the minimum value of f

(C) $\int_0^{\pi/2} y\,dx = 3 - \frac{\pi}{2}$ (D) $f(\pi/3) = 1$

Ans. C, D

Sol. Differentiating both sides, we have

$f'(x) - \sin 2x = -f(x)\tan x$

For $y = f(x)$, we have $y' + y\tan x = \sin 2x$

This is a linear equation whose integration factor is $e^{\int \tan x\,dx} = \sec x$. The general solution is

$y(x) = \cos x\left[\int \sin x\,dx + C\right] = C\cos x - 2\cos^2 x$

Since y(0) = f(0)=1, so C = 3. Thus

$f(x) = y = 3\cos x - 2\cos^2 x = -2(\cos x - 3/4)^2 + \frac{9}{8}$

$\int_0^{\pi/2} y\,dx = \int_0^{\pi/2} (3\cos x - (1+\cos 2x))dx = 3 - \frac{\pi}{2}$

and $f(\pi/3) = \frac{3}{2} - \frac{1}{2} = 1$

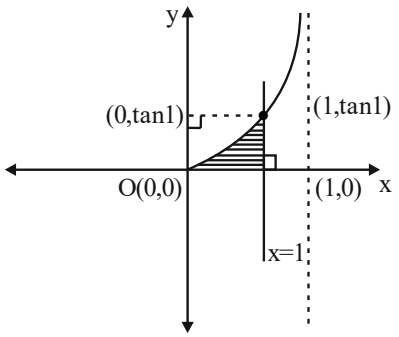
12. Area of the region bounded by the curve $y = \tan x$ and lines y = 0 and x = 1 is equal to

(A) $\int_0^1 \tan(1-x)dx$ (B) $\int_0^{\tan 1} \tan^{-1}y\,dy$

(C) $\int_0^1 \tan^{-1}x\,dx$ (D) $\tan 1 - \int_0^{\tan 1} \tan^{-1}x\,dx$

Ans. A, D

Sol.



Required area $= \int_0^1 \tan x\,dx = \int_0^1 \tan(1-x)dx$ (Using King property)

Also, required area $= \tan 1 - \int_0^{\tan 1} \left(\tan^{-1} y\right)dy$

13. Let $f:[0,1]\to R$ be continuous and differentiable function such that for any $x,y\in[0,1]$ $xf(y)+yf(x)\le 1$, then which of the following options is/are correct?

(A) $\int_0^1 f(x)dx \le \frac{\pi}{4}$ (B) $\int_0^1 x^2 f'(x)dx \ge f(1)-1$

(C) $\int_0^1 xf'(x)dx \ge f(1)-\frac{\pi}{4}$ (D) $\int_0^1 f(x)dx \le \frac{\pi}{8}$

Ans. A, B, C

Sol. Let $x=\sin\theta$

$dx=\cos\theta\,d\theta$ & $I=\int_0^1 xf'(x)dx=\int_0^{\frac{\pi}{2}} f(\cos\theta)\sin\theta\,d\theta$

$2I\le\int_0^{\frac{\pi}{2}} d\theta \Rightarrow I\le\frac{\pi}{4}$ equality holds is $f(x)=\sqrt{1-x^2}$

now, $x=y\Rightarrow xf(x)\le\frac{1}{2}$

$\int_0^1 f(x)dx\le\int_0^1\frac{1}{2}dx$

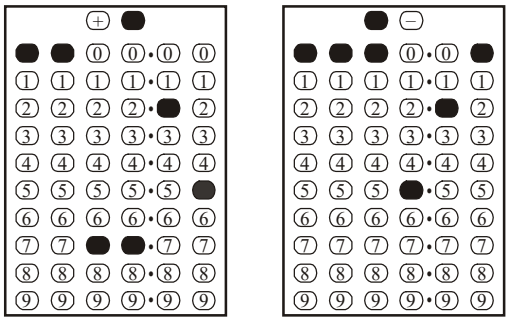
$\left(\frac{x^2}{2}f(x)\right)_0^1-\int_0^1\frac{x^2}{2}f(x)dx\le\frac{1}{2}$

$f(x)\int_0^1 x^2f^2(x)dx\le 1$

also $\int_0^1 f(x)dx\le\frac{\pi}{4}$ (Using by parts)

SECTION-II : (Maximum Marks: 20)

- This section contains **FIVE (05)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.
  For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +4 If ONLY the correct numerical value is entered as answer.
  Zero Marks : 0 In all other cases.

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14. The parabola $P: y = ax^2$ where 'a' is a positive real constant, is touched by the line $L: y = mx - b$ (where m is a positive constant and b is real) at the point T. Let Q be the point of intersection of the line L and the y-axis such that TQ = 1. If A denotes the maximum value of the area of the region surrounded by P, L and the x-axis, find the value of $\frac{1}{A}$.

Ans. 12

Sol. $y = ax^2$

$$\left.\frac{dy}{dt}\right|_T = 2ax_0 = m$$

Hence, line is

$$y = (2ax_0)x - b \qquad ...(i)$$

$\therefore (x_0, ax_0^2)$ lies on parabola and the line (i)

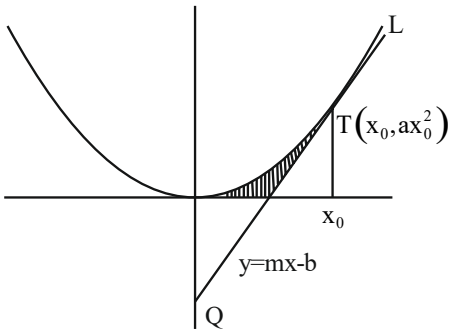
$$ax_0^2 = 2ax_0^2 - b$$

$b = ax_0^2$, hence $Q = (0,-b) = (0,-ax_0^2)$

Now using $(TQ)^2 = 1$

$$x_0^2 + 4a^2x_0^4 = 1$$

$$a^2 = \frac{(1 - x_0^2)}{4x_0^4} \qquad ...(ii)$$



Now, $A=\int_0^{x_0}\left(ax^2-mx+b\right)dx=\frac{ax^3}{3}-\frac{mx^2}{2}+bx\Big|_0^{x_0}=\frac{ax_0^3}{3}-\frac{mx_0^2}{2}+bx_0$

$=\frac{ax_0^3}{3}-ax_0^3+ax_0^3=\frac{ax_0^3}{3}$

$\therefore A^2=\frac{a^2x_0^6}{9}=\frac{x_0^6}{9}\left(\frac{1-x_0^2}{4x_0^4}\right)=\frac{x_0^2\left(1-x_0^2\right)}{36}$

Let $A^2=f\left(x_0\right)=\frac{x_0^2\left(1-x_0^2\right)}{36}$

This is maximum when $x_0^2=\frac{1}{2}$

$A^2\big|_{max}=\frac{1}{2}.\frac{1}{2}.\frac{1}{36}=\frac{1}{144};$

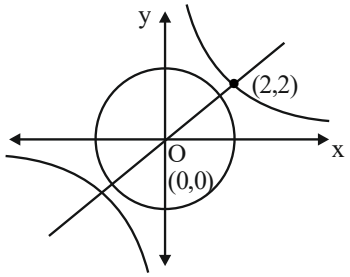
$\therefore A_{max}=\frac{1}{12}\Rightarrow\frac{1}{A}=12$

15. Let $y=f\left(x\right)$ be a curve C_1 passing through (2, 2) and $\left(8,\frac{1}{2}\right)$ and satisfying a differential equation $y\left(\frac{d^2y}{dx^2}\right)=2\left(\frac{dy}{dx}\right)^2$. Curve C_2 is the director circle of the circle $x^2+y^2=2$. If the shortest distance between the curves C_1 and C_2 is $\left(\sqrt{p}-q\right)$ where $p,q\in N$, then find the value of $\left(p^2-q\right)$.

Ans. 62

Sol. Given $y\left(\frac{d^2y}{dx^2}\right)=2\left(\frac{dy}{dx}\right)^2\Rightarrow\frac{y''}{y'}=\frac{2y'}{y}\Rightarrow\ln y'=2\ln y+\ln a$

$\Rightarrow\frac{y'}{y^2}=a\Rightarrow\int\frac{dy}{y^2}=\int a\,dx\Rightarrow\frac{-1}{y}=ax+b$



Since curve is passing through (2, 2) and $\left(8,\frac{1}{2}\right)$

So, $2a+b=\frac{-1}{2}$ ...(i)

$8a+b=-2$ ...(ii)

$\therefore$ On solving (i) and (ii), we get $a=\frac{-1}{4}, b=0$

Hence, $C_1: xy=4$ and curve C_2 is $x^2+y^2=4$

$\therefore$ Shortest distance between C_1 and $C_2 = 2\sqrt{2}-2=\sqrt{8}-2$

Hence , $\left(p^2-q\right)=64-2=62$

16. Find the number of values of x satisfying $\int_0^x t^2\sin(x-t)dt=x^2$ in [0, 100].

Ans. 16

Sol. Given $\int_0^x t^2\sin(x-t)dt=x^2$

$\Rightarrow \int_0^x (x-t)^2\sin t\,dt=x^2$

$\Rightarrow x^2\int_0^x \sin t\,dt-2x\int_0^x t\sin t+\int_0^x t^2\sin t=x^2$

$\Rightarrow x^2(1-\cos x)-2x(-x\cos x+\sin x)+\left(-x^2\cos x+2x\sin x+\cos x-1\right)=x^2$

$\Rightarrow \cos x=1$

So, $x=2n\pi,\ n\in I$

Hence, number of values of x = $0,2\pi,4\pi,.....,30\pi$ equals 16

17. Let $P(x)$ be a polynomial with real coefficients such that

$\left(x^2+x+1\right)P(x-1)=\left(x^2-x+1\right)P(x)\ \forall x\in R$ and $P(1)=3$.

If $\int_0^1 \tan^{-1}\left(\frac{2x}{1+P\left(x^2\right)}\right)dx+\int_0^1 \tan^{-1}(x+1)dx=\frac{k}{16}(\pi-\ln 4)$, then find the value of k.

Ans. 12

Sol. $\left(x^2+x+1\right)P\left(x-1\right)=\left(x^2-x+1\right)P\left(x\right)$ ..(i)

Since, x^2+x+1 and x^2-x+1 have no factor

Let $p\left(x\right)=\left(x^2+x+1\right)g\left(x\right)$

from (i), $g\left(x-1\right)=g\left(x\right)$

$\Rightarrow g\left(x\right)=g\left(x+1\right)\Rightarrow$ periodic

A polynomial is periodic if it is constant,

$\therefore g\left(x\right)=k$

$\Rightarrow P\left(x\right)=k\left(x^2+x+1\right),\ P\left(1\right)=3$

$\Rightarrow P\left(x\right)=x^2+x+1$

Now, $\int_0^1\tan^{-1}\left(\frac{2x}{1+P\left(x^2\right)}\right)dx=\int_0^1\tan^{-1}\left(\frac{2x}{1+x^4+x^2+1}\right)dx$

$=\int_0^1\tan^{-1}\left(\frac{2x}{1+\left(x^2+x+1\right)\left(x^2-x+1\right)}\right)dx$

$=\int_0^1\left(\tan^{-1}\left(x^2+x+1\right)-\tan^{-1}\left(x^2-x+1\right)\right)dx$

$=\int_0^1\left(\frac{\pi}{2}-\cot^{-1}\left(x^2+x+1\right)-\frac{\pi}{2}+\cot^{-1}\left(x^2-x+1\right)\right)dx$

$=\int_0^1\left(\cot^{-1}\left(x^2-x+1\right)-\cot^{-1}\left(x^2+x+1\right)\right)dx$

$=\int_0^1\left(\tan^{-1}\left(\frac{1}{1+x^2-x}\right)-\tan^{-1}\left(\frac{1}{1+x^2+x}\right)\right)dx$

$=\int_0^1\left(\tan^{-1}x-\tan^{-1}\left(x-1\right)-\tan^{-1}\left(1+x\right)+\tan^{-1}x\right)dx$

$=\int_0^1\tan^{-1}x\,dx+\int_0^1\tan^{-1}x\,dx-\int_0^1\tan^{-1}\left(x+1\right)+\int_0^1\tan^{-1}x$

$\int_0^1\tan^{-1}\left(\frac{2x}{1+P\left(x^2\right)}\right)dx+\int_0^1\tan^{-1}\left(1+x\right)dx$

$=3\int_0^1\tan^{-1}x\,dx$

$=3\left[\tan^{-1}x.x\Big|_0^1-\int_0^1\frac{x}{1+x^2}\right]$

$=3\left[\frac{\pi}{4}-\frac{1}{2}\ln 2\right]=\frac{3}{4}\left[\pi-2\ln 2\right]$

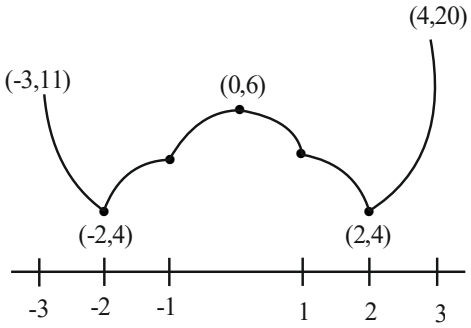
$=\frac{3}{4}\left[\pi-\ln 4\right]$

$k=12$

18. Sum of all integral value(s) of k for which $|x-1|+|x^2-4|+|x+1|=k$ has 2 distinct real solutions in the interval $x \in [-3,4]$ is

Ans. 49

Sol.



For two solutions $k=(6,11]\cup\{4\}$

$\therefore$ Sum of all values = 49

ENTHUSIAST SRG TEST SERIES

TEST - 8 (PAPER 2)	DATE : 10 - 10 - 2021

SYLLABUS

PHYSICS	:	Electromagnetic Induction, Alternating Current, Em Waves, Simple Harmonic Motion
CHEMISTRY	:	Aldehyde and ketone, Carboxylic acid and its derivative, Thermodynamics
MATHS	:	Definite Integration, Area Under The Curve, Differential Equations, Quadratic Equations

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

path to success

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

TARGET : JEE (M + A) : 2022

ENTHUSIAST SRG TEST SERIES

TEST TYPE: ONLINE | **DATE : 10.10.2021** | **PATTERN : JEE ADVANCED**

Time : 3 Hours | Maximum Marks : 216

TEST - 08 (PAPER 2)

PART 1 - PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8		
	A.	AB	BD	AC	AC	AB	BC	AC	D		
SECTION-IV	Q.	9	A	B	C	D	10	A	B	C	D
			R	S	Q	P		S	R	P	Q
SECTION-II	Q.	11	12	13	14	15	16	17	18		
	A.	3	2	1	6	6	5	2	6		

PART 2 - CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8		
	A.	AC	BC	BC	ABD	BC	ABCD	ABC	CD		
SECTION-IV	Q.	9	A	B	C	D	10	A	B	C	D
			QRS	PQRT	QS	QR		PST	PRT	QS	PR
SECTION-II	Q.	11	12	13	14	15	16	17	18		
	A.	4	2	4	7	4	6	1	1		

PART 3 - MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8		
	A.	ABCD	CD	BD	CD	ABD	BC	ABC	CD		
SECTION-IV	Q.	9	A	B	C	D	10	A	B	C	D
			P,Q	R,S	P	R,S		R	P	S	Q
SECTION-II	Q.	11	12	13	14	15	16	17	18		
	A.	5	2	5	13	6	8	7	6		

PART-1 : PHYSICS

SECTION–I : (Maximum Marks : 32)

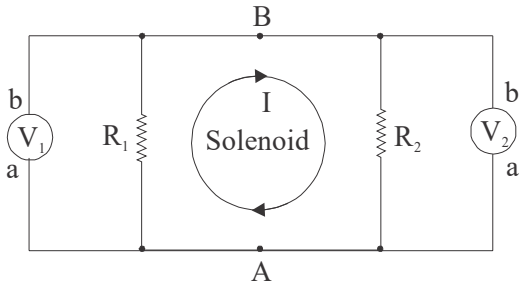
- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –1	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

---

1. The current in a long solenoid is increasing linearly with time, so the flux is proportional to $t : \Phi = \alpha t$. Two voltmeters are connected to diametrically opposite points (A and B), together with resistors $(R_1$ and $R_2)$, as shown in fig. Assume that these are ideal voltmeters that draw negligible current (they have huge internal resistance), and that a voltmeter registers $-\int_a^b E \cdot d\ell$ between the terminals and through the meter.



(A) $V_1 = \dfrac{\alpha R_1}{(R_1 + R_2)}$

(B) $V_2 = -\dfrac{\alpha R_2}{(R_1 + R_2)}$

(C) $V_1 = V_2 = \alpha$

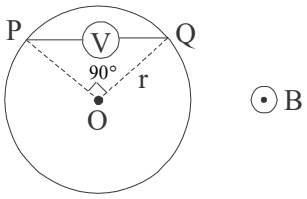
(D) $V_1 = V_2 = \dfrac{\alpha R_1 R_2}{R_1 + R_2}$

Ans. AB

Sol. $\varepsilon = -\dfrac{d\Phi}{dt} = -\alpha$. So the current in R_1 and R_2 is $I = \dfrac{\alpha}{R_1 + R_2}$; by Lenz's law, it flows counterclockwise.

Now the voltage across R_1 (which voltmeter $\neq 1$ measures) is $V_1 = IR_1 = \dfrac{\alpha R_1}{R_1 + R_2}$ (V_b is the higher potential), and $V_2 = -IR_2 = \dfrac{-\alpha R_2}{R_1 + R_2}$ (V_b is lower).

2. A circular wire loop (radius r, resistance R) encloses a region of uniform magnetic field, B, perpendicular to its plane. The field increases linearly with time $(B = \alpha t)$. An ideal voltmeter (infinite internal resistance) is connected between points P and Q.



(A) The current in the loop is $\frac{\pi\alpha r^2}{2R}$

(B) The reading of voltmeter is $\frac{\alpha r^2}{2}$

(C) The reading of voltmeter is αr^2

(D) The current in loop is $\frac{\pi\alpha r^2}{R}$

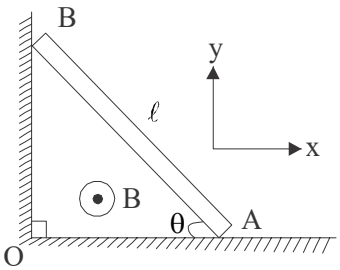
Ans. BD

Sol. (a) $\varepsilon = -\frac{d\Phi}{dt} = -\pi r^2 \frac{dB}{dt} = -\alpha\pi r^2 = IR \Rightarrow$ (in magnitude) $I = \frac{\pi\alpha r^2}{R}$. If B is out of the page, Lenz's law says the current is clockwise.

(b) In ΔOPQ $|\varepsilon| = \frac{d\phi}{dt} = A\frac{dB}{dt} = \left(\frac{1}{2}\times r^2\right)\alpha$.

Hence, reading of voltmeter is $\frac{1}{2}r^2\alpha$

3. A conducting rod of length ℓ slides in a vertical plane against two insulated perpendicular surfaces as shown. Find the induced emf across the rod of its end A is pulled with $\vec{v} = v_0\hat{i}$. Outward uniform magnetic field of induction B is applied.



(A) $\varepsilon = \frac{B\ell V_0}{2}$ when $\theta = 30°$

(B) $\varepsilon = B\ell V_0$ when $\theta = 30°$

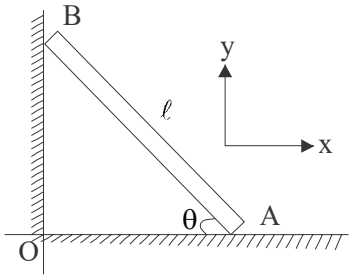
(C) $\varepsilon = 0$ when $\theta = 45°$

(D) $\varepsilon = \frac{B\ell V_0}{\sqrt{2}}$ when $\theta = 45°$

Ans. AC

Sol. Assume ABO as the loop at any instant t. The area enclosed by the loop $= A = \frac{1}{2}(\ell\cos\theta)(\ell\sin\theta)$

$= \frac{\ell^2}{4}\sin 2\theta$. Hence, the flux ϕ through the loop is given as $\phi = \vec{B}.\vec{A} = \left(B\hat{k}\right)\cdot\left(\frac{\ell^2}{4}\sin 2\theta\,\hat{k}\right) = \frac{B\ell^2}{4}\sin 2\theta$



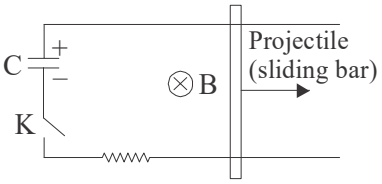
Differentiating ϕ w.r.t. time t we obtain $\varepsilon_m = -\frac{d\phi}{dt} = -\frac{B\ell^2}{4}(2\cos 2\theta)\frac{d\theta}{dt}$,

Where $\frac{d\theta}{dt} = \frac{v}{\ell \sin\theta}$

Then, $\varepsilon = \frac{B\ell v_0}{2} = \frac{\cos 2\theta}{\sin\theta}$

4. Given figure is a device to fire (project) a body (sliding bar) with a great speed in the order of km/s by an Ampere's force on the conducting sliding bar offered by an impulsive current sent by the charged capacitor after closing the key in the inward magnetic field.

If Q_0 is the initial charge of the capacitor C, M = mass of the sliding bar and B = magnetic field (produced by the closely spaced parallel rails assumed to be uniform), answer the following questions: (at t = 0, K is closed)



(A) The final charge on capacitor $\frac{B^2\ell^2CQ_0}{M + CB^2\ell^2}$

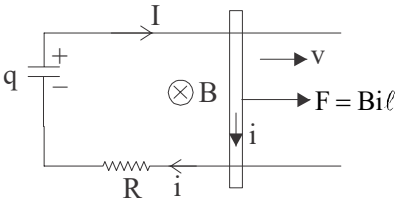
(B) The maximum (muzzle) speed of the projectile is $\frac{2B\ell Q_0}{M + CB^2\ell^2}$

(C) The maximum (muzzle) speed of the projectile is $\frac{B\ell Q_0}{M + CB^2\ell^2}$

(D) The final charge on capacitor $\frac{2B^2\ell^2CQ_0}{M + CB^2\ell^2}$

Ans. AC

Sol.



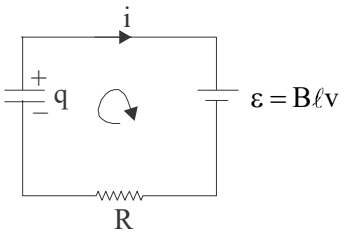
Force on rod

$Bi\ell = Ma$

$$B\left(-\frac{dq}{dt}\right)\ell = M\frac{dv}{dt}$$

$$\Rightarrow B\ell(Q_0 - q) = MV \quad ---(1)$$

Equivalent circuit diagram



$$\frac{q}{c} - B\ell v - iR = 0 \qquad ---(2)$$

$$\frac{q}{c} - \frac{B^2\ell^2}{M}(Q_0 - q) == iR$$

For final change $i = 0 \Rightarrow \frac{q}{c} = \frac{B^2\ell^2}{M}(Q_0 - q)$

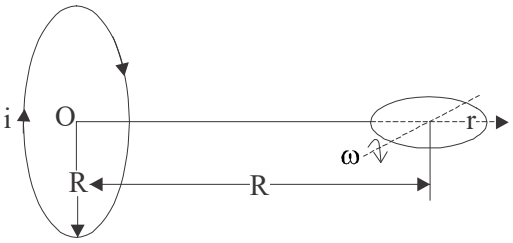
$$\Rightarrow q\left(1 + \frac{B^2 + \ell^2 C}{M}\right) = \frac{B^2\ell^2 CQ_0}{M}$$

$$\Rightarrow q_{final} = \frac{B^2\ell^2 CQ_0}{M + B^2L^2C}$$

From eq 2

$$V_{max} = \frac{q_{final}}{B\ell C} = \frac{B\ell Q_0}{M + B^2\ell^2 C}$$

5. A small loop of radius $r(<< R)$, where R = radius of the larger loop that carries a current i is placed on the axis of the bigger loop. If the smaller loop is rotated through $90°$, then the (the resistance of the small loop is R_0). [ω = Angular velocity of loop along the diametrical axis]



(A) Change in flux is $\frac{\pi\mu_0 r^2 i}{4\sqrt{2}R}$

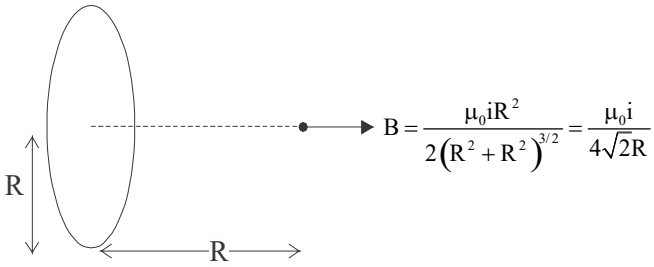
(B) Average current flowing through smaller loop is $\frac{\mu_0 r^2 \omega i}{2\sqrt{2}R_0 R}$

(C) Change in flux is $\frac{\pi\mu_0 r^2 i}{2\sqrt{2}R}$

(D) Average current flowing through smaller loop is $\frac{\mu_0 r^2 \omega i}{\sqrt{2}R_0 R}$

Ans. AB

Sol.



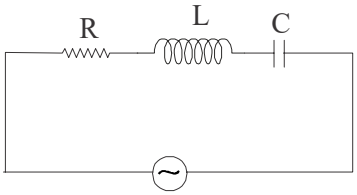
$$B=\frac{\mu_0 iR^2}{2\left(R^2+R^2\right)^{3/2}}=\frac{\mu_0 i}{4\sqrt{2}R}$$

$$\Delta\phi=B(\text{Area})=\frac{\mu_0 i\pi r^2}{4\sqrt{2}R}$$

$$\text{Average current}=\frac{\Delta q}{\Delta t}=\frac{\Delta\phi}{R_0}\times\frac{1}{T/4}=\frac{\Delta\phi}{R_0}\left(\frac{2\omega}{\pi}\right)$$

$$i_{ave}=\frac{\mu_0 ir^2\omega}{2\sqrt{2}RR_0}$$

6.



$$V=220\sqrt{2}\sin(100\pi t)$$

$$R=100\Omega,\ L=\frac{2}{\pi}H,\ C=\frac{1}{10\pi}mF$$

In above given AC circuit, V_R, V_L and V_C are potential drop across resistance, inductor and capacitor at an instant t. Then

(A) $V_C=220\sin(100\pi t-\pi/4)$ (B) $V_L=440\sin(100\pi t+\pi/4)$

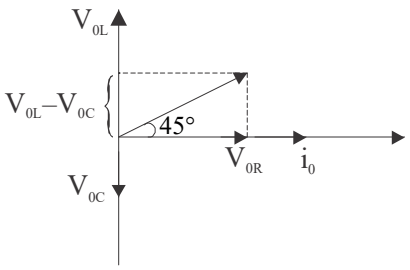
(C) $V_C=220\sin(100\pi t-3\pi/4)$ (D) $V_L=220\sin(100\pi t-\pi/4)$

Ans. BC

Sol. $Z=\sqrt{R^2+(x_L-x_C)^2}=\sqrt{100^2+100^2}=100\sqrt{2}$

$$i_0=\frac{V_0}{Z}=\frac{200\sqrt{2}}{100\sqrt{2}}=2.2A$$

$$\text{Power factor}=\cos\phi=\frac{R}{2}=\frac{1}{\sqrt{2}}\Rightarrow\phi=45°$$



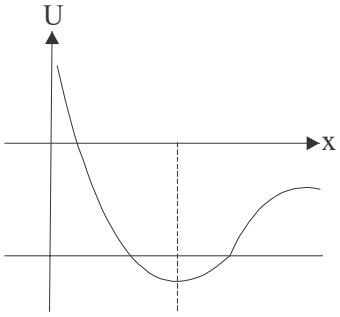
So circuit is inductive

So, $i = i_0 \sin(100\pi t - \pi/4) = 2.2\sin(100\pi t - \pi/4)$

$V_L = (200 \times 2.2)\sin(100\pi t + \pi/4) = 440\sin(100\pi t + \pi/4)$

$V_C = (100 \times 2.2)\sin(100\pi t - 3\pi/4) = 220\sin(100\pi t - 3\pi/4)$

7. The potential energy between two atoms in a diatomic molecule varies with x as $U = \frac{a}{x^{12}} - \frac{b}{x^6}$, where a and b are positive constants. Consider oscillation of one atom if the other atom is kept stationary.



(A) Equivalent spring constant $= \left(\frac{9b^2}{a}\right)\left(\frac{b}{2a}\right)^{1/3}$ (B) Equivalent spring constant $= \left(\frac{9b^2}{a}\right)\left(\frac{2b}{a}\right)^{1/3}$

(C) Equilibrium position of particle $= \left(\frac{2a}{b}\right)^{1/6}$ (D) Equivalent position of particle $= \left(\frac{a}{2b}\right)^{1/6}$

Ans. AC

Sol. Differentiating $U = \frac{a}{x^{12}} - \frac{b}{x^6}$ with respect to x, we have $\frac{dU}{dx} = \frac{-12a}{x^{13}} + \frac{6b}{x^7}$

Since $F = \frac{-dU}{dx} = 0$ for the particle to stay in equilibrium, the equilibrium point $x = x_0$ can be given as

$\frac{-12a}{x_0^{13}} + \frac{6b}{x_0^7} = 0$

This gives $x_0 = \sqrt[6]{\frac{2a}{b}}$

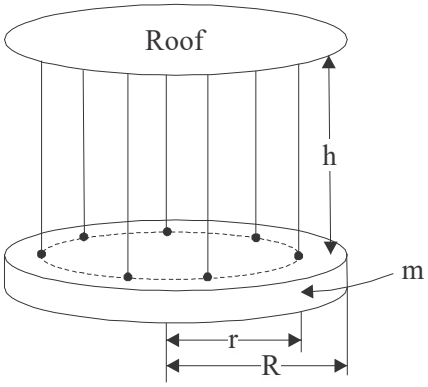
Differentiating $\frac{dU}{dx}$ with respect to x, we have, $\frac{d^2U}{dx^2} = -12 \times -13\frac{a}{x^{14}} + \frac{(6b)(-7)}{x^8}$

Evaluating $k_{eff} = \frac{d^2U}{dx^2}$ at $x = x_0 = \sqrt[6]{\frac{2a}{b}}$,

We have $k_{eff} = \frac{+156a}{\left(\sqrt[6]{\frac{2a}{b}}\right)^{14}} - \frac{42b}{\left(\sqrt[6]{\frac{2a}{b}}\right)^{8}}$

This gives $k_{eff} = \frac{9b^2}{a}\left(\frac{b}{2a}\right)^{1/3}$

8. A uniform disc of mass m and radius R is hanging from a rigid support by N–number of strings each of length h as shown in the figure. Find the time period of small torsional oscillation about vertical axis passing through the CM of the disc.



(A) $\pi\left(\frac{R}{r}\right)\sqrt{\frac{h}{g}}$

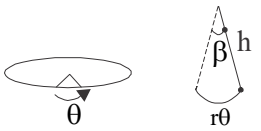
(B) $2\pi\left(\frac{R}{r}\right)\sqrt{\frac{2h}{g}}$

(C) $2\pi(R/r)\sqrt{\frac{h}{g}}$

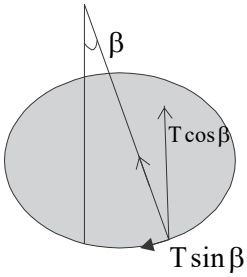
(D) $\pi(R/r)\sqrt{\frac{2h}{g}}$

Ans. D

Sol. When disc is rotated by angle θ, string rotated by β.



$h\beta = r\theta$



At equilibrium $NT = mg$

$T = mg/N$

Torque about axis

$$N(T\sin\beta)\times r = \left(\frac{mR^2}{2}\right)\alpha$$

$$mgr\times\left(\frac{r\theta}{h}\right) = \left(\frac{mR^2}{2}\right)\left(\frac{d\theta}{dt^2}\right)$$

$$\frac{d^2\theta}{dt^2} = -\left(\frac{2gr^2}{hR^2}\right)\theta$$

$$\Rightarrow \omega^2 = \left(\frac{2gr^2}{hR^2}\right)$$

$$T = 2\pi\sqrt{\frac{hR^2}{2gr^2}} = \pi(R/r)\sqrt{\frac{2h}{g}}$$

SECTION–IV : (Maximum Marks : 16)

- This section contains **TWO** questions.
- Each question contains two columns, **Column-I** and **Column-II.**
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **column-II.**
- One or more entries in **Column-I** may match with one or more entries in **Column-II.**
- The ORS contains a 4 × 5 matrix whose layout will be similar to the one shown below :

(A) (P) (Q) (R) (S) (T)
(B) (P) (Q) (R) (S) (T)
(C) (P) (Q) (R) (S) (T)
(D) (P) (Q) (R) (S) (T)

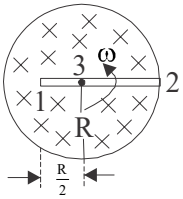
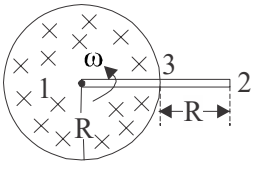
- For each entry in **column-I,** darken the bubbles of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then darken these three bubbles in the ORS. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

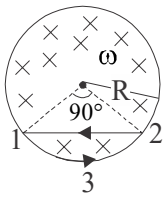
For each entry in **Column-I**

Full Marks	: +2	If only the bubble(s) corresponding to all the correct match(es) is (are) darkened
Zero Marks	: 0	In none of the bubbles is darkened
Negative Marks	: 0	In all other cases

9. In column – I time varying magnetic field is available in circular region of radius R. $R = 1m, \frac{dB}{dt} = 2T/s$

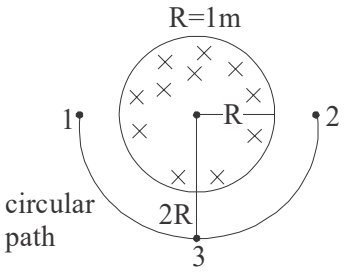
In column – II induced emf or $\oint \vec{E}.\vec{d\ell}$ value is given. Match ε value in column – II with different situations in column – I $(\omega = 2rad/sec$ and $B = 1T)$.

Column I Conductor in a time varying magnetic field	**Column II** Induced emf
(A)  ε = P.D. between points 1 and 2	(P) $\varepsilon = 3.14$ volt
(B)  ε = P.D. between points 1 and 2	(Q) $\varepsilon = 0.57$ volt

(C) 

(R) $\varepsilon = 3/4$ volt

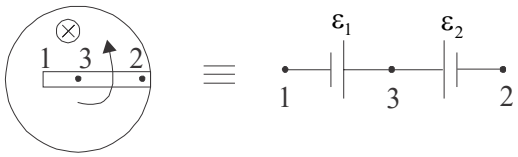
$\varepsilon = \oint \vec{E}.\overrightarrow{d\ell}$ in loop 1321

(D) 

(S) $\varepsilon = 1$ volt

$\varepsilon = \int \vec{E}.\overrightarrow{d\ell}$ in circular path 132 of radius 2R

Ans. A – R, B – S, C – Q, D – P

Sol. (A) 

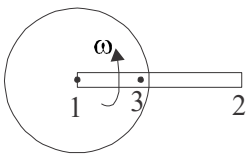
$$\varepsilon_1 = \frac{Bw(R/2)^2}{2} = \frac{B\omega R^2}{8}$$

$$\varepsilon_2 = \frac{Bw(R)^2}{2}$$

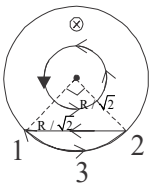
$$\varepsilon_{132} = +\varepsilon_1 - \varepsilon_2$$

$$= \left(\frac{1}{8} - \frac{1}{2}\right) B\omega R^2 = -\frac{3}{8} \times 1 \times 2 \times 1^2$$

$$|\varepsilon_{132}| = \frac{3}{4} V$$

(B) 

$$\varepsilon_{132} = \frac{B\omega R^2}{2} = \frac{1 \times 2 \times 1^2}{2} = 1$$

(C) 

$$\frac{dB}{dt} = 2T/s$$

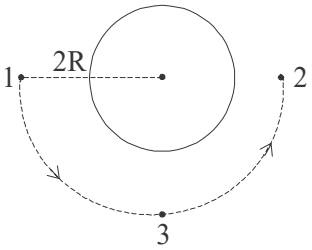
E = Induced electric field

$$\varepsilon_{132} = E \times 2\pi R \times \frac{1}{4} = \left(\frac{R}{2}\right)\left(\frac{dB}{dt}\right)(\pi R/2)$$

$$\varepsilon_{132} = \frac{1}{2} \times 2 \times \pi \times \frac{1}{2} = \pi/2 = 1.57\,V$$

$$\varepsilon_{12} = \frac{1}{2} \times \left(\sqrt{2}R\right)\left(R/\sqrt{2}\right) = R^2/2 \times \frac{dB}{dt} = 2 \times \frac{1}{2} = 1$$

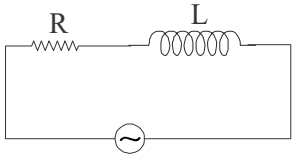
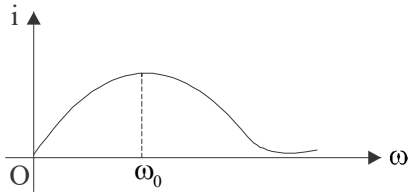
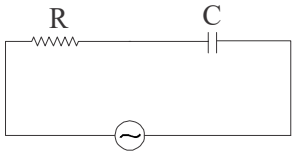
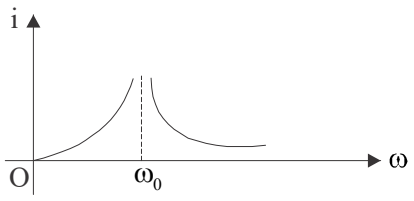
$$\int_{132} \vec{E} \cdot \vec{d\ell} = 1.57 - 1 = 0.57\,V$$

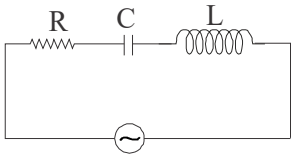
(D) 

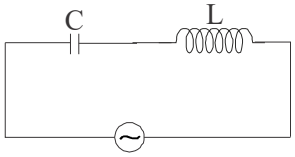
$$\varepsilon_{132} = \frac{dB}{dt} \times \frac{\pi R^2}{2}$$

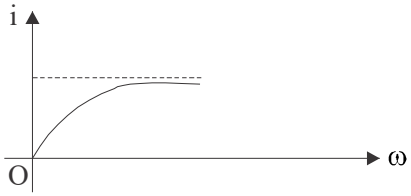
$$\varepsilon_{132} = 2 \times \frac{1}{2} \times \pi \times 1^2 = \pi$$

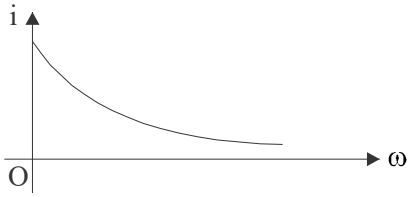
10. In column – I AC circuits are given and in column – II variation of i with respect to ω is give. Match the following. Value of R,L and C is constant and only ω of source is varied.

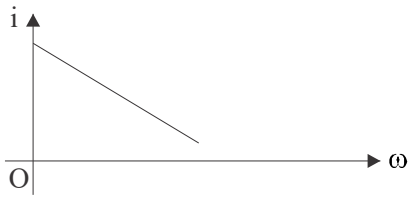
Column I	**Column II**
A.C.Circuit	$i = f(\omega)$
(A)  $V = V_0 \sin(\omega t)$	(P) 
(B) 	(Q) 

(C) 

(D) 

(R) 

(S) 

(T) 

Ans. A – S, B – R, C – P, D – Q

Sol. (A) $Z=\sqrt{R^2+X_L^2}$ $\quad i_0=\dfrac{V_0}{Z}=\dfrac{V_0}{\sqrt{R^2+X_L^2}}$

$\omega=0,\ i=\dfrac{V_0}{R}$, then decreases

(B) $Z=\sqrt{R^2+\left(\dfrac{1}{\omega c}\right)^2}$, $\omega=0, Z=\infty, i=0$ as $\omega\uparrow$ $i\uparrow$ $\omega\uparrow$ $i\uparrow$

(C) RLC series circuit

(D) $Z=\left|X_L-X_C\right|$ $\quad i=\dfrac{V}{\left(X_L-X_C\right)}$

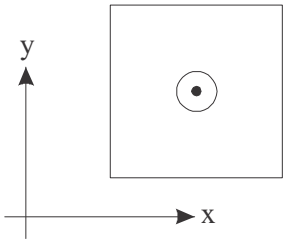
at $\omega=\omega_0=\dfrac{1}{\sqrt{LC}}$, $Z=0, i=\infty$

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

11. A square loop of side $\ell = 10$ cm with its sides parallel to x and y axes moves with velocity $v = -7\hat{i}$ m/s. An outward magentic field is present which decreases along negative x–axis with a gradient $\frac{\partial B}{\partial x} = 0.1 T/m$ and increases with time at a rate of $\frac{\partial B}{\partial t} = 0.1 T/s$. If the resistance of the loop is 2×10^{-3} ohms, then find the magnitude and direction of induced current in the loop. (in Ampere)



Ans. 3

Sol. Direct application of the fomula yields $\frac{d\phi}{dt} = \left(\frac{\partial\phi}{\partial t} + \frac{\partial\phi}{\partial x}\frac{dx}{dt}\right) = -\varepsilon$

Put $\frac{\partial\phi}{\partial t} = \frac{\partial B}{\partial t}\cdot A = \left(\ell^2\right) = \frac{\partial B}{\partial t} = (0.1)^2 \times (0.1) = 10^{-3}$ wb / s

and $\frac{\partial\phi}{\partial t}\cdot\frac{\partial x}{\partial t} = \ell^2 \cdot \left(\frac{\partial B}{\partial x}\right)\cdot\left(\frac{dx}{dt}\right) = (0.1)^2 \times (0.1) \times -7$

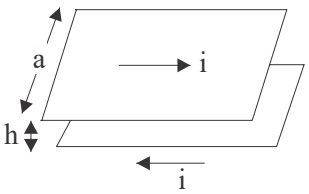
Summing up both terms obtain total induced emf

$$\varepsilon = -\frac{d\phi}{dt}$$

$$= -\left(10^{-3} - 7\times10^{-3}\right) = +6\times10^{-3}\ V$$

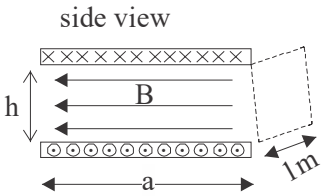
$$i = \frac{\varepsilon}{R} = \frac{6\times10^{-3}}{2\times10^{-3}} = 3\,A$$

12. Two thin long conducting tapes of which width "a" are separated through a distance "h" as shown in the figure. Current a is uniformly distributed over (the area of) each tape. Assuming linear density of the current over its width uniform the inductance of the tape per its unit length is $\frac{\mu_0 h^x}{a^y}$. Find value of $x + y$.



Ans. 2

Sol.



$$B=\left(\frac{\mu_0\lambda}{2}\right)\times 2=\mu_0\lambda=\left(\frac{\mu_0 i}{a}\right)$$

$$B=\frac{\mu_0 i}{a}$$

$$\phi = B\times \text{ Area}$$

$$=\mu_0\times\frac{i}{a}(h\times 1)=\left(\frac{\mu_0 h}{a}\right)i=Li$$

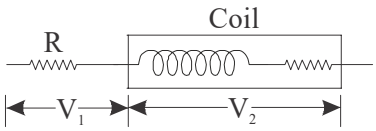
$$\text{So } L=\frac{\mu_0 h}{a}$$

value of x + y = 2

13. A resistor R is connected in series with a coil (Resistance not negligible). The system is subjected to an A.C. Supply of peak voltage V_0. If the peak voltages across the resistor R and the coils are V_1 and V_2 respectively, find the power (in watt) dissipated in the coil. (assume $V_1=5V,\ V_2=7V, V_0=10V,\ R=13\Omega$)

Ans. 1

Sol.

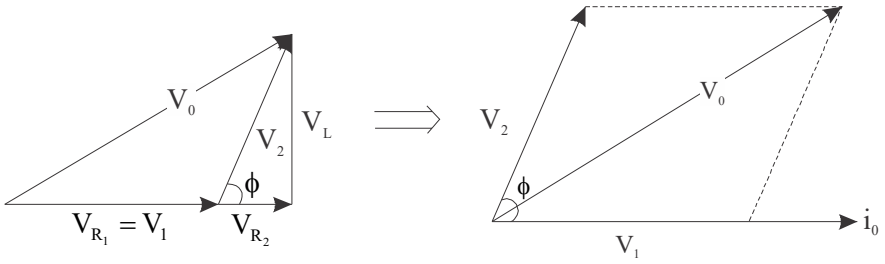


The power dissipated in the coil is $P=\frac{i_0 V_2}{2}\cos\phi \quad ---(i)$

$$=\frac{1}{2}\left(\frac{V_1}{R}\right)V_2\cdot\cos\phi$$

From the phasor diagram, $V_1^2+V_2^2+2V_1\times V_2\cos\phi=V_0^2 \quad ---(ii)$

Substituting $V_2\cos\phi$ from eq. (ii) in eq. (i),



$$P=\frac{1}{2R}\left(\frac{V_0^2-V_1^2-V_2^2}{2}\right)=\frac{1}{4R}\left[V_0^2-V_1^2-V_2^2\right]$$

$$P=\frac{1}{2\times13}\left(10^2-5^2-7^2\right)=1\,W$$

14. In R–L–C series circuit, we have same current at angular frequencies ω_1 and ω_2. Find the resonant frequency. (Take $\omega_1=4\,units,\ \omega_2=9\,units$)

Ans. 6

Sol. The current in series R–L–C circuit is

$$i=\frac{V_0}{\sqrt{R^2+\left|\omega L-\frac{1}{\omega C}\right|^2}}$$

Since, $i_1=i_2$ at ω_1 and ω_2, we have

$$\text{or,}\quad \frac{V_0}{\sqrt{R^2+\left(\omega_1L-\frac{1}{\omega_1C}\right)^2}}=\frac{V_0}{\sqrt{R^2+\left(\omega_2L-\frac{1}{\omega_2C}\right)^2}}$$

$$\text{or,}\quad \omega_1L-\frac{1}{\omega_1C}=\omega_2L-\frac{1}{\omega_2C}$$

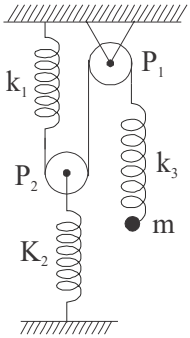
$$\text{or,}\quad L\left(\omega_1-\omega_2\right)=\frac{1}{C}\left(\frac{1}{\omega_1}-\frac{1}{\omega_2}\right)$$

$$\text{or,}\quad \omega_1\omega_2=\frac{1}{LC}$$

Substituting $\frac{1}{LC}=\omega_0^2$, we have

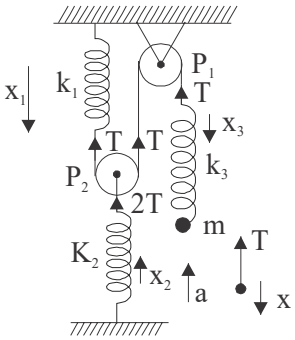
$$\omega_0=\sqrt{\omega_1\omega_2}=\sqrt{4\times9}=2\times3=6\,units$$

15. A body of mass m hangs from a smooth fixed pully P_1 by the inextensible string fitted with the springs of stiffnesses k_1 and k_3. The string passes over the smooth light pully P_2 which is connected with another ideal spring of stiffness k_2. Find the period of oscillation of the body. (in seconds) assume all springs masslass and take $k_1=k_3=\frac{k_2}{4}=\pi^2$ SI units , $M=3\,kg$



Ans. 6

Sol.



Let assume M downwards by x, due to which springs elongated by x_1, x_2 and x_3 respectively.

T = change in tension force $x_1 + 2x^2 + x_3 = x \quad -----(1)$

and $T = k_1x_1 = k_3x_3 = \dfrac{k_2x_2}{2} \quad -----(2)$

$T = Ma \quad ---(3)$

From eq (1) & (2) $\Rightarrow \dfrac{T}{k_1} + \dfrac{4T}{k_2} + \dfrac{T}{k_3} = x$

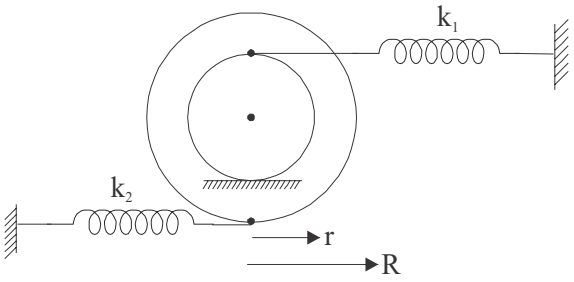
$$T = \left(\frac{1}{\frac{1}{k_1} + \frac{4}{k_2} + \frac{1}{k_3}}\right)x = Ma \quad \Rightarrow a = -\left\{\frac{1}{M\left(\frac{1}{k_1} + \frac{4}{k_2} + \frac{1}{k_3}\right)}\right\}x$$

Comparing with SHM

$$\omega^2 = \frac{1}{M\left(\frac{1}{k_1} + \frac{4}{k_2} + \frac{1}{k_3}\right)}$$

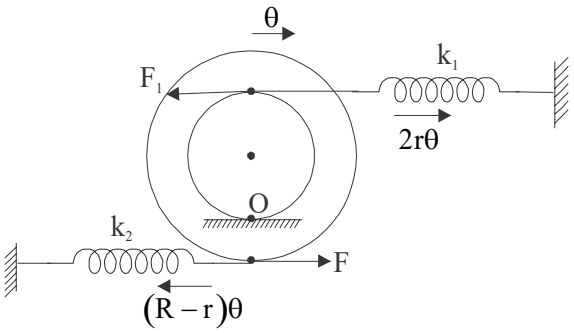
Time period $= 2\pi\sqrt{M\left(\dfrac{1}{k_1} + \dfrac{4}{k_2} + \dfrac{1}{k_3}\right)} = \dfrac{2\pi}{\pi}\sqrt{3\times 3} = 6\,\text{sec}$

16. A stepped pully having mass m, radius R and radius of gyration about cm is connected with two ideal springs of stiffness k_1 and k_2 as shown in the figure. If the pulley rolls without sliding, the angular frequency of its oscillation is given by $\sqrt{\dfrac{xk_1r^2 + yk_2(R-r)^2}{M(K^2+r^2)}}$. Find value of x + y.



Ans. 5

Sol.



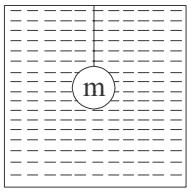
Torque about O point $\tau_0 = F_1(2r) + F_1(R-r)$

$$M\left(K^2 + r^2\right)\alpha = \left[K_1(2r)^2 + K_2(R-r)^2\right]\theta$$

$$\alpha = \left\{\frac{4K_1r^2 + K_2(R-r)^2}{M\left(K^2+r^2\right)}\right\}\theta$$

$x + y = 4 + 1 = 5$

17. A simple pendulum has time period T_0, say. When it is placed in a liquid the time period decreases. Then we need to move the point of suspension with a vertical acceleration a (a = 2g) to regain the time period of simple pendulum. The ratio of densities of liquid and the bob is:



Ans. 2

Sol. $T_0 = 2\pi\sqrt{\ell / g}$,

When placed in liquid of density $= \rho$

$$T^1 = 2\pi\sqrt{\frac{\ell}{g\left(1-\frac{\rho}{\sigma}\right)}} \qquad \sigma = \text{ density of loop}$$

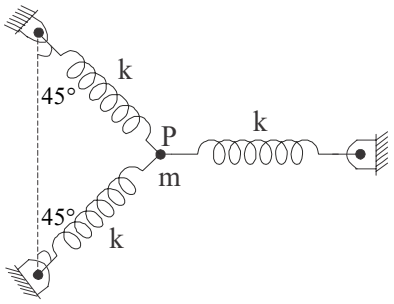
Now accelerated upward

$$T^{11} = 2\pi\sqrt{\frac{\ell}{g\left(1-\frac{\rho}{\sigma}\right)+a}} = T.$$

$$\Rightarrow a = g\left(\frac{\rho}{\sigma}\right)$$

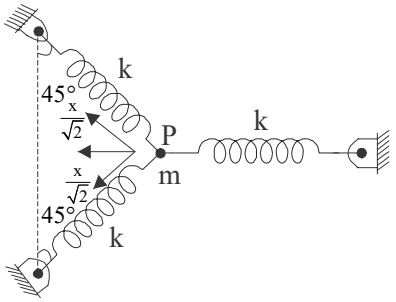
$$\Rightarrow \frac{\rho}{\sigma} = \frac{a}{g} = 2$$

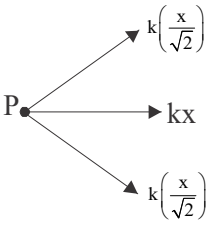
18. Period of oscillation of the particle P is $2\pi\sqrt{\frac{3m}{xk}}$, then x is equal to



Ans. 6

Sol. Lets assume m moves left by x.





$$F_{net} = \frac{kx}{\sqrt{2}} \times \left(\frac{1}{\sqrt{2}}\right) \times 2 + kx = 2kx = ma$$

$$a = \left(\frac{2k}{m}\right)x$$

$$T = 2\pi\sqrt{\frac{m}{2k}}$$

$$\Rightarrow T = 2\pi\sqrt{\frac{3m}{6k}}$$

$$x = 6$$

PART-2 : CHEMISTRY

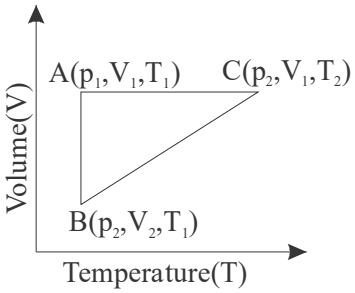
SECTION–I : (Maximum Marks : 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –1	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

1. A reversible cyclic process for an ideal gas is shown below. Here, p, V and T are pressure, volume and temperature, respectively. The thermodynamic parameters q, w, H and U are heat, work, enthalpy and internal energy, respectively.



The correct options is (are)

(A) $q_{AC} = \Delta U_{BC}$ and $W_{AB} = nRT_1 \ell n \frac{V_1}{V_2}$

(B) $W_{BC} = P_2(V_1 - V_2)$ and $W_{AC} = 0$

(C) $\Delta H_{CA} < \Delta U_{AC}$ and $q_{BC} = \Delta H_{AC}$

(D) $q_{AC} = \Delta U_{BC}$ and $\Delta H_{CA} > \Delta U_{CA}$

Ans. AC

Sol. In the given curve AC represents isochoric process as volume at both the points is same i.e., V_1

Similarly, AB represents isothermal process (as both the points are at T_1 temperature) and BC represents isobaric process as both the points are at P_2 pressure.

Nor (i) for option (a)

$q_{AC} = \Delta U_{BC} = nC_V(T_2 - T_1)$

Where, n = number of moles

C_v = specific heat capacity at constant volume

However, $W_{AB} \neq p_2(V_2 - V_1)$ instead

$$|W_{AB}| = nRT_1 \ln\left(\frac{V_2}{V_1}\right)$$

$q_{BC} = \Delta H_{AC} = nC_p(T_2 - T_1)$

Where, C_p = specific heat capacity at constant pressure likewise,

$W_{BC} = -P_2(V_1 - V_2)$

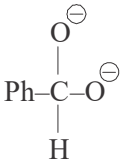
as $nC_p(T_2 - T_1) < nC_V(T_2 - T_1)$

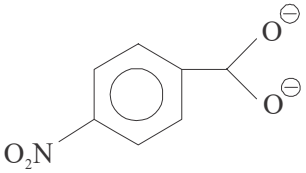
so $\Delta H_{CA} < \Delta U_{CA}$

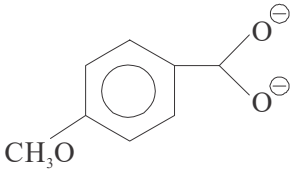
and $q_{AC} = \Delta U_{BC}$

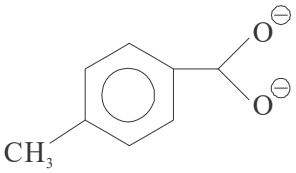
Although $q_{BC} = \Delta H_{AC}$

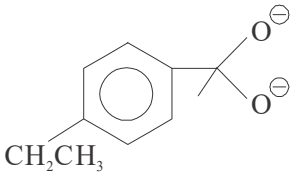
but $\Delta H_{CA} \not< \Delta U_{CA}$

2. No.of compound which are better hydride donor than 

(A) 

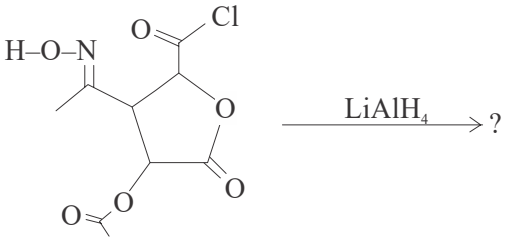
(B) 

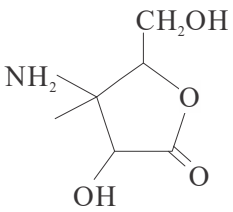
(C) 

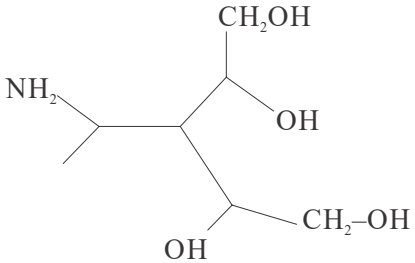
(D) 

Ans. BC

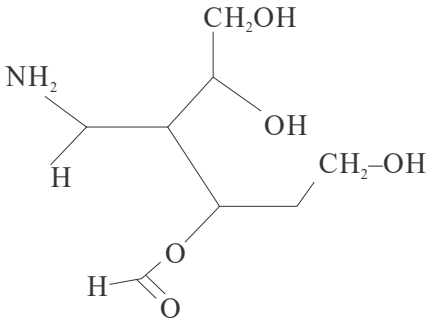
3. Find out the possible products in the following reaction.



(A) 

(B) 

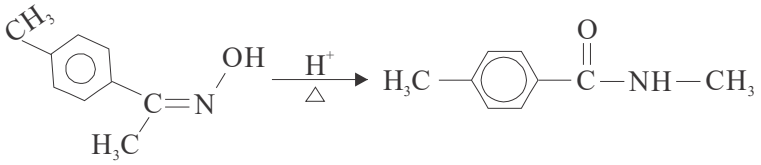
(C) CH_3CH_2OH

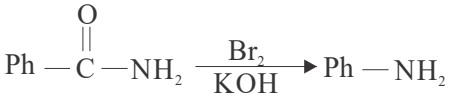
(D) 

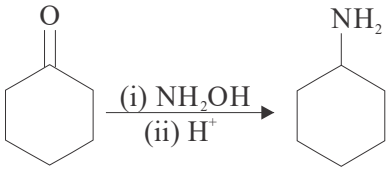
Ans. BC

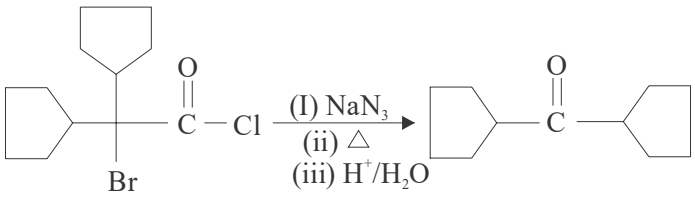
Sol.

4. Which of the following reactions represent the major product?

(A) 

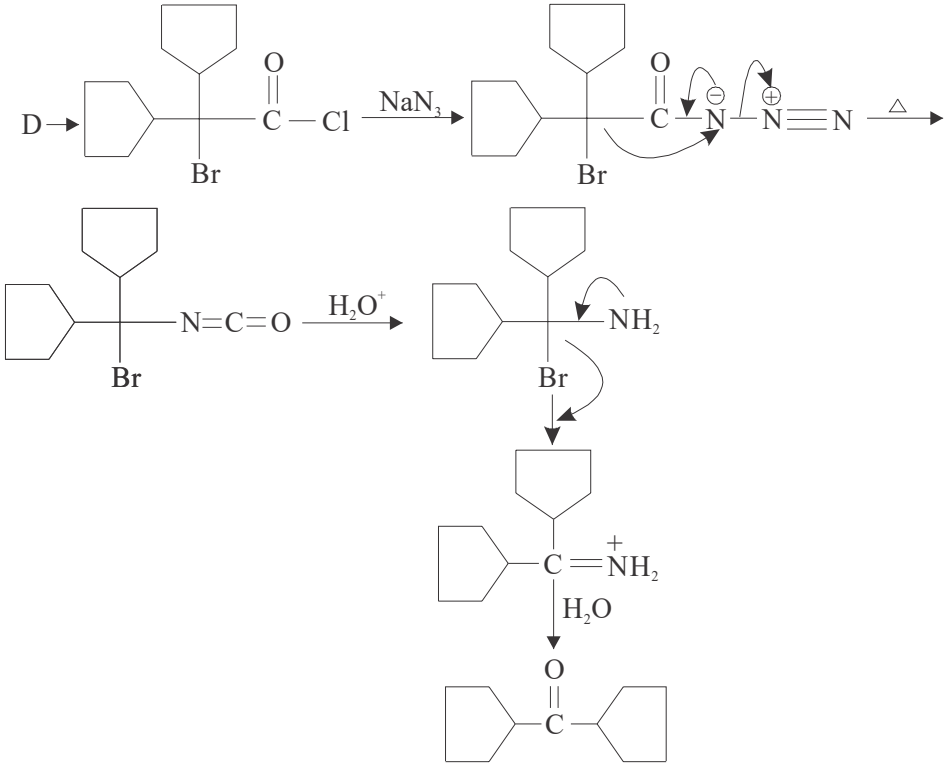
(B) 

(C) 

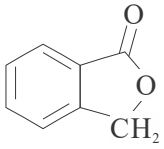
(D) 

Ans. ABD

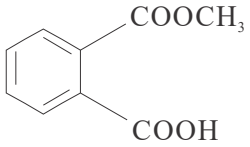
Sol. A is Beckmann rearrangement

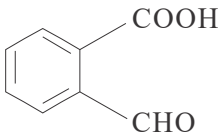


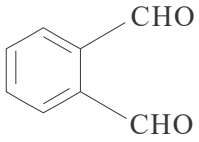
5. Which of the following reactants on reaction with conc. NaOH followed by acidification gives following lactone as the main product?

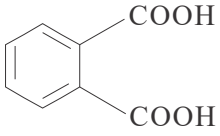


lactone

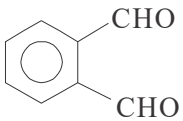
(A) 

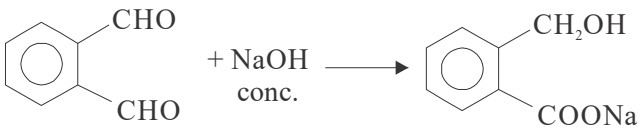
(B) 

(C) 

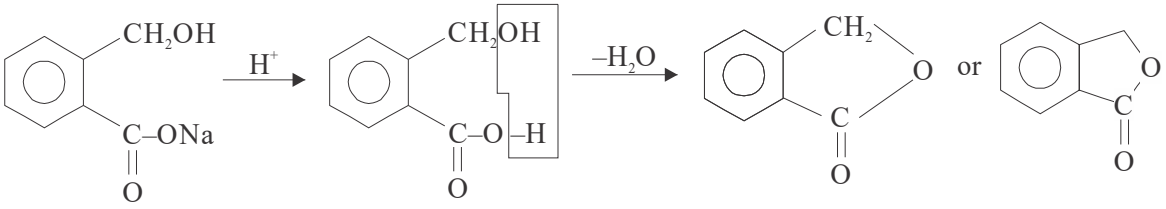
(D) 

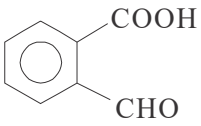
Ans. BC

Sol.  gives Cannizaro's reaction with conc. NaOH

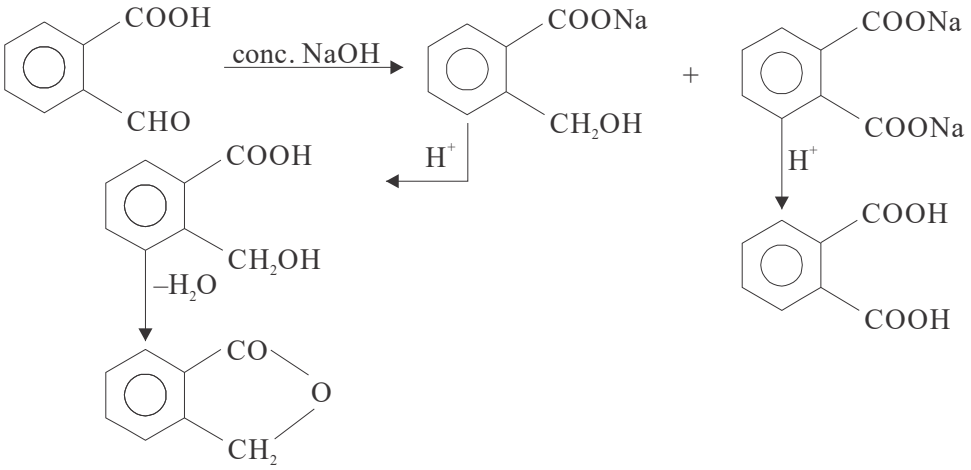


This compound on acidification gives a compound which have $-CH_2OH, -COOH$ groups gives lactone on dehydration.





also undergo Cannizaro's reaction with conc. NaOH



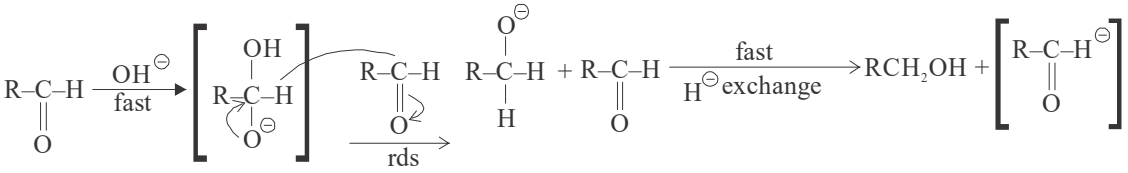
6. Which of the following products is/are correctly mentioned in the following reactions.

(A) $HCOH \xrightarrow{NaOD} HCOONa + CH_3OD$ (B) $HCDO \xrightarrow{NaOH} DCOONa + CH_2DOH$

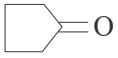
(C) $HCDO \xrightarrow{NaOD} DCOONa + DCH_2OD$ (D) $D_2CO \xrightarrow{NaOD} DCOONa + CD_3OD$

Ans. ABCD

Sol. Cannizaro reaction

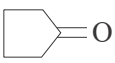


7. $C_4H_6O_4(X) \xrightarrow{LAH} A \xrightarrow{PCl_3} B \xrightarrow{NaCN} C \xrightarrow{H_3O^+, CaO, \Delta} Y \xleftarrow{CaO, \Delta}$ hexanedioic acid

(A) B is $CH_2(CH_2Cl)-CH_2(CH_2Cl)$ (B) Y is 

(C) X is $CH_2(COOH)-CH_2(COOH)$ (D) C is $CH_2(CH_2COOH)-CH_2(CH_2COOH)$

Ans. ABC

Sol. $\xrightarrow{CaO, \Delta}$ 

8. A sample containing 1.0 mol of an ideal gas is expanded isothermally and reversibly to ten times of its original volume, in two separate experiments. The expansion is carried out at 300 K and at 600 K respectively. Choose the correct option.

(A) Work done at 600 K is 20 times the work done at 300 K

(B) Work done at 300 K is twice the work done at 600 K

(C) Work done at 600 K is twice the work done at 300 K

(D) $\Delta U = 0$ in both cases

Ans. CD

Sol. Justification : $\frac{W_{600K}}{W_{300K}} = \frac{1 \times R \times 600 \ \ln\frac{10}{1}}{1 \times R \times 300 \ \ln\frac{10}{1}} = \frac{600}{300} = 2$

For isothermal expansion of ideal gases, $\Delta U = 0$

Since temperature is constant this means there is no change in thermal energy.

Therefore, $\Delta U = 0$

SECTION–IV : (Maximum Marks : 16)

- This section contains **TWO** questions.
- Each question contains two columns, **Column-I** and **Column-II.**
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **column-II.**
- One or more entries in **Column-I** may match with one or more entries in **Column-II.**
- The ORS contains a 4 × 5 matrix whose layout will be similar to the one shown below :

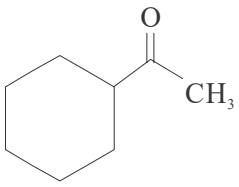
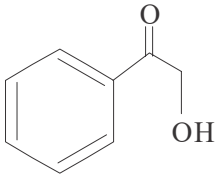
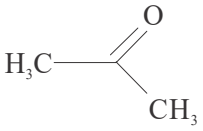
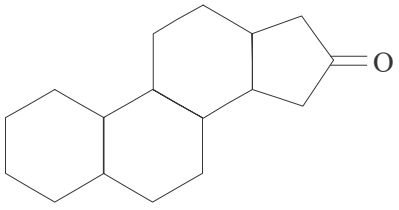
(A) (P) (Q) (R) (S) (T)
(B) (P) (Q) (R) (S) (T)
(C) (P) (Q) (R) (S) (T)
(D) (P) (Q) (R) (S) (T)

- For each entry in **column-I,** darken the bubbles of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then darken these three bubbles in the ORS. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks	: +2	If only the bubble(s) corresponding to all the correct match(es) is (are) darkened
Zero Marks	: 0	In none of the bubbles is darkened
Negative Marks	: 0	In all other cases

9. Match the compounds/ions in Column I with their properties/reactions in Column II.

Column I	**Column II**
(A) 	(P) Gives red precipitate with $CuSO_4$ / sodium potassium tartrate solution
(B) 	(Q) Shows tautomerism
(C) 	(R) Gives two isomeric compounds with hydrazine under acidic condition
(D) 	(S) Gives yellow precipitate with I_2 / NaOH
	(T) Gives silver mirror test with tollens reagent

Ans. A–QRS ; B–PQRT ; C–QS ; D–QR

Sol. Factual

10. Match the following

Column I	Column II
(A) Isothermal reversible expansion	(P) $\Delta S_{system} > 0$
(B) Isothermal irreversible	(Q) $\Delta S_{system} = 0$
(C) Adiabatic reversible expansion	(R) $\Delta S_{universe} > 0$
(D) Adiabatic irreversible expansion	(S) $\Delta S_{universe} = 0$
	(T) $\Delta S_{surrounding} < 0$

Ans. A–PST; B–PRT; C–QS; D–PR

Sol. Factual

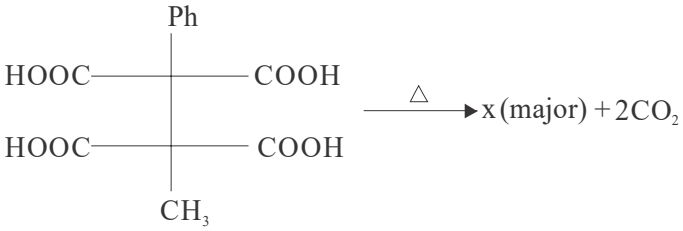
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.

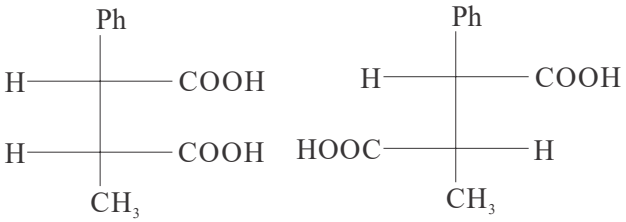
Zero Marks : 0 In all other cases.

11. The number of products possible for x (including stereomers) in the following reaction is:



Ans. 4

Sol.



and their images

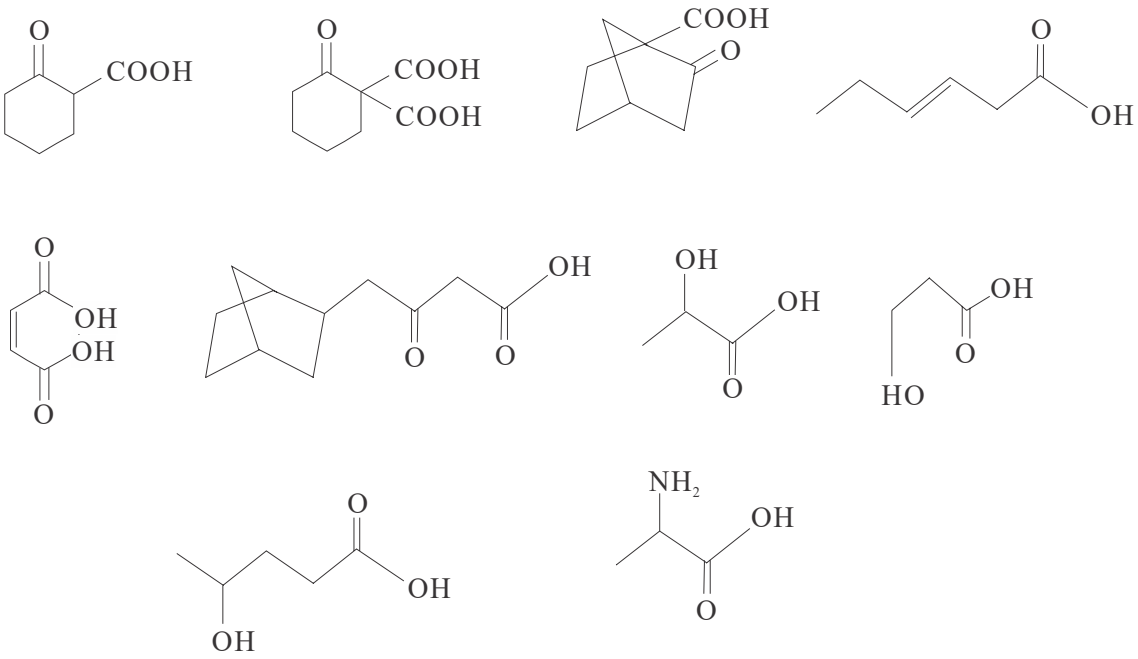
12. Which of the reactants have maximum rate of reaction as compared to Benzaldehyde for the formation of cyanohydrin compound

1. p–Methyoxy benzaldehyde
2. p–Nitrobenzaldehyde
3. p–Methylbenzaldehyde
4. p–aminobenaldehyde
5. p–Hydroxybenzaldehyde
6. p–Anilinium benzaldehyde

Ans. 2

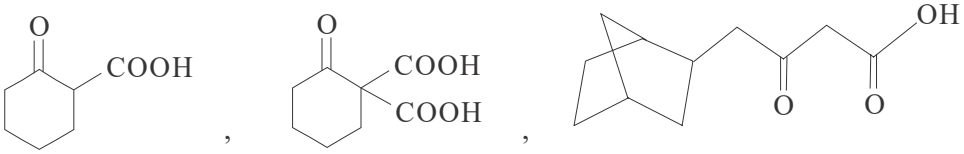
Sol. Strong deactivating group at the para position of the benzene ring increases the electrophilicity of carbonyl carbon.

13. How many of the following compounds can undergo decarboxylation on heating.

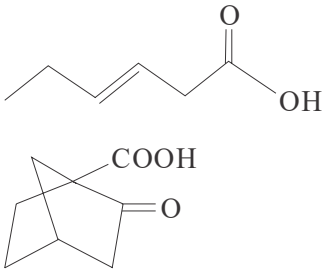


Ans. 4

Sol. β – keto acids an heaing undergo decarboxylaty



β – δ unsaturated carboxylic acids also undergo decarboxylation on heating

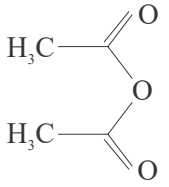


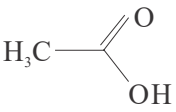
Even though it is a β – keto acid does not undergo decarboxylation on heating the transition state is unstable.

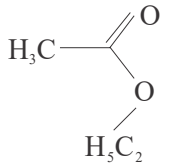
14. An organic compound 'A' on treatment with ethyl alcohol gives a carboxylic acid 'B' and compound 'C'. Hydrolysis of 'C' under acidic conditions gives 'B' and 'D'. Oxidation of 'D' with $KMnO_4$ also gives 'B'. 'B' on heating with $Ca(OH_2)$ gives 'E' (C_3H_6O). E does not give Tollen's test and does not reduce Fehling's solution but form a 2,4–dinitrophenyl hydrazone. Let no.of carbons are present in compound 'A' is 'x' and oxygen be 'y', then $x + y = ?$

Ans. 7

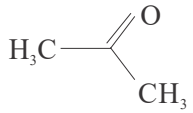
Sol. $x = 4$; $y = 3$

A = 

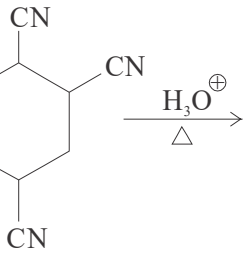
B = 

C = 

D = H_5C_2–OH

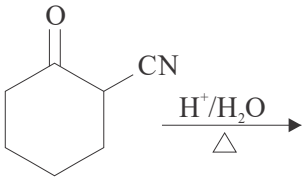
E = 

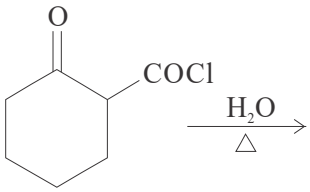
15. Number of reactions in which product has one carbon lesser than reactant

(I) 

(II) $CH_3CONH_2 + OBr^- \xrightarrow[\Delta]{H_3O^{\oplus}}$

(III) $CH_3COOH + N_3H + H_2SO_4 \xrightarrow{\Delta}$

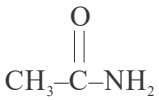
(IV) 

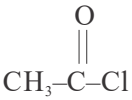
(V) 

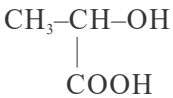
Ans. 4

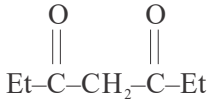
Sol. II, III, IV and V

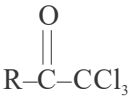
16. Which of the following will show '+' ve haloform test?

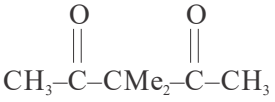
(1) 

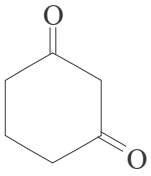
(2) 

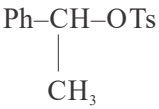
(3) 

(4) 

(5) 

(6) 

(7) 

(8) 

Ans. 6

Sol. 3,4,5,6,7,8

17. Isothermal compressibility (K) of an ideal gas is defined as $K = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_{T,n}$. What is the isothermal compressibility factor (per atm) for an ideal gas at 1.0 atm?

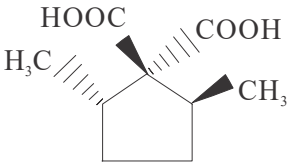
Ans. 1

Sol. We know that at constant T & n, $PV = C$ (constant) [According to Boyle's law]

$$V = \frac{C}{P} \text{ or } \left(\frac{\partial V}{\partial P}\right)_{T,n} = -\frac{C}{P^2} = -\frac{V}{P}\left[\because V = \frac{C}{P}\right]$$

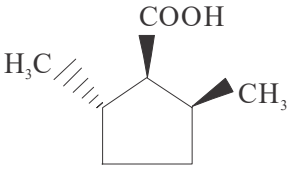
$$\therefore K = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_{T,n} = \frac{1}{P} = 1 atm^{-1}$$

18. Find the total number of products obtained when



Undergoes decarboxylation

Ans. 1

Sol. 

PART-3 : MATHEMATICS

SECTION–I : (Maximum Marks : 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –1	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

---

1. If $f(x)=\lim\limits_{m\to\infty}\dfrac{1}{m}\sum\limits_{k=0}^{m}\left|x-\dfrac{k}{m}\right|$; then

(A) $f(0)=\dfrac{1}{2}$ (B) $f(1)=\dfrac{1}{2}$

(C) $f(-1)=\dfrac{3}{2}$ (D) $f(2)=\dfrac{3}{2}$

Ans. ABCD

Sol. $f(x)=\int\limits_0^1|x-t|dt=\begin{cases}\int\limits_0^1(t-x)dt, x\le 0\\ \int\limits_0^x(x-t)dt+\int\limits_x^1(t-x)dt, o<x<\\ \int\limits_0^1(x-t)dt, x\ge 1\end{cases}$

$$\Rightarrow f(x)=\begin{cases}\dfrac{1-2x}{2}, x\le 0\\ \dfrac{2x^2-2x+1}{2}, 0<x<1\\ \dfrac{2x-1}{2}, x\ge 1\end{cases}$$

2. Let f(x) is a differentiable function such that $f'(x)=f(x)+\int_0^2 f(x)dx$ if $f(0)=\frac{4-e^2}{3}$, then

(A) $f(x)=e^x-\left(\frac{e^2-1}{3}\right)$ (B) $f(x)=e^x+\left(\frac{e^2-1}{3}\right)$

(C) $f(0)<0$ (D) f(x) has atleast one root between 0 & 1

Ans. CD

Sol. $f(x)=\int_0^2 f(x)\,dx$

Let $\int_0^2 f(x)\,dx=K\Rightarrow f'(x)-f(x)=K\Rightarrow e^{-x}f(x)=k\int e^{-x}dx\Rightarrow f(x)=ce^x-k$

Since $\int_0^2 f(x)\,dx=k\Rightarrow k=\frac{c(e^2-1)}{3}$

& $f(0)=\frac{4-e^2}{3}\Rightarrow c=1,k=\frac{e^2-1}{3}$

$f(x)=e^x-\left(\frac{e^2-1}{3}\right)$

3. The differential equation $\{xy^3(1+x\cos x)-y\}dx+xdy=0$ represents the curve

$\frac{x^2}{ay^2}=\frac{x^3}{b}+x^2\sin x+cx\cos x-d\sin x+k.$ Then

(A) $c+d=0$ (B) $a+b+c+d=9$

(C) $b+d=-1$ (D) $a+b=5$

Ans. BD

Sol. $\frac{dy}{dx}+y^3(1+\cos x)-\frac{y}{x}=0$

$\frac{1}{y^3}\frac{dy}{dx}-\frac{1}{y^2x}=-(1+\cos x)$

Let $\frac{-1}{y^2}=u$

$\therefore\frac{1}{2}\frac{du}{dx}+\frac{u}{x}=-(1+\cos x)$

$IF=e^{\int\frac{2}{x}dx}=x^2$

$\therefore ux^2=-2\int x^2(1+\cos x)dx$

$\Rightarrow\frac{x^2}{2y^2}=\frac{x^3}{3}+x^2\sin x+2x\cos x-2\sin x+k$

4. Solution of equation $y\,dx\left(1+x^2\cdot\ln xy\right)+x\,dy\left(1+y^2\ln\left(xy\right)\right)=0$ is

(A) $\ln\left(\ln xy\right)+x^2+y^2+c=0$

(B) $\ln xy=ce^{\frac{\left(x^2+y^2\right)}{2}}$

(C) $2\ln\left(\ln xy\right)+x^2+y^2+c=0$

(D) $\ln xy=ce^{-\frac{\left(x^2+y^2\right)}{2}}$

Ans. CD

Sol. $x\,dy+y\,dx+\ln xy\left(x^2y\,dx+xy^2\,dy\right)=0$

$$\frac{1}{xy}\left(xdy+ydx\right)+\ln\left(xy\right)\left(xdy+ydx\right)=0$$

$$\Rightarrow d\left(\ln xy\right)+\frac{1}{2}\left(xy\right)d\left(x^2+y^2\right)=0$$

$$\Rightarrow 2\ln\left|\left(\ln xy\right)\right|+x^2+y^2+c=0$$

$$\therefore\ln\left(\ln xy\right)=-\frac{\left(x^2+y^2+c\right)}{2}$$

$$\ln xy=ce^{-\left(\frac{x^2+y^2}{2}\right)}$$

5. If $ax^2+bx+c=0$ and $cx^2+bx+a=0$ $\left(a,b,c\in R\right)$ have a common non-real root, then

(A) $-2|a|<b<2|a|$

(B) $-2|c|<|b|<2|c|$

(C) $a=\pm c$

(D) $a=c$

Ans. ABD

Sol. $D_1=b^2-4ac<0, D_2=b^2-4ac<0,$ as the root is non–real

$$\Rightarrow\frac{a}{c}=\frac{b}{b}=\frac{c}{a}=1\Rightarrow a=c$$

Now, $b^2-4ac<0\Rightarrow b^2-4a^2\left(\text{or}\,4c^2\right)<0$

$$\Rightarrow|b|<2|a|\left(\text{or}\ 2|c|\right)$$

6. The equation $a_8x^8+a_7x^7+a_6x^6+.....+a_0=0$ has all its roots positive and real (where $a_8=1, a_7=-4, a_0=1/2^8$), then

(A) $a_1=\frac{1}{2^8}$

(B) $a_1=-\frac{1}{2^4}$

(C) $a_2=\frac{7}{2^4}$

(D) $a_3=\frac{1}{2^2}$

Ans. BC

Sol. Let the roots be $\alpha_1,\alpha_2,.....,\alpha_8$

$$\Rightarrow\alpha_1+\alpha_2+.....+\alpha_8=4$$

$\alpha_1 \cdot \alpha_2 .....\alpha_8 = \dfrac{1}{2^8}$

$\Rightarrow (\alpha_1 \cdot \alpha_2 .....\alpha_8)^{1/8} = \dfrac{1}{2} = \left(\dfrac{\alpha_1 + \alpha_2 + ... + \alpha_8}{8}\right)$

$\Rightarrow$ A.M = G.M. $\Rightarrow$ all the roots are equal to $\dfrac{1}{2}$

$\Rightarrow a_1 = -{}^8C_7\left(\dfrac{1}{2}\right)^7 = -\dfrac{1}{2^4}$

$a_2 = {}^8C_6\left(\dfrac{1}{2}\right)^6 = \dfrac{7}{2^4}$

$a_3 = -{}^8C_5\left(\dfrac{1}{2}\right)^5$

7. A differentiable function satisfies $f(x) = \int_0^x \{f(t)\cos t - \cos(t-x)\}dt$. Which is of the following hold good?

(A) f(x) has a minimum value 1–e

(B) f(x) has a maximum value $1 - e^{-1}$

(C) $f''\left(\dfrac{\pi}{2}\right) = e$

(D) $f'(0) = 1$

Ans. ABC

Sol. $f(x) = \int_0^x \{f(t)\cos t - \cos(t-x)\}dx$

$f(x) = \int_0^x f(t)\cos t\, dt - \int_0^x \cos(t-x)dx \qquad \left[\int_0^a f(x)dx = \int_0^a f(\alpha - x)dx\right]$

$f(x) = \int_0^x f(t)\cos t\, dt - \sin x$

Differentiating both the sides, we get

$f'(x) = f(x)\cos x - \cos x$

Let $f(x) = y;\ f'(x) = \dfrac{dy}{dx}$

$\dfrac{dy}{dx} - y\cos x = -\cos x$

$IF = e^{-\int \cos x\, dx} - e^{-\sin x}$

Therefore, $y \cdot e^{-\sin x} = -\int e^{-\sin x}\cos x\, dx;\ y = Ce^{\sin x} + 1$

If x = 0; y = 0 (from the given relation)

$\Rightarrow C = -1$

Therefore, $f(x) = 1 - e^{\sin x}$

Now, minimum value $=1-e$ $\left(\text{When } x=\pi/2\right)$

Maximum value $=1-e^{-1}$ $\left(\text{When } x=-\pi/2\right)$

$f'(x)=-e^{\sin x}\cos x$

Therefore, $f'(0)=-1$

$f''(x)=-\left[\cos^2 x\cdot e^{\sin x}-e^{\sin x}\cdot\sin x\right]$

$f''\left(\frac{\pi}{2}\right)=e$

8. A curve $y=f(x)$ passes through $(1,1)$ and tangent at $P(x,y)$ cuts the x–axis at A and B respectively such that $BP:AP=3:1$, then

(A) Equation of curve is $x^3y+1=0$

(B) Tangent at $(1,1)$ is $x+3y=4$

(C) Curve passes through $\left(3,\frac{1}{27}\right)$

(D) Equation of curve is $x^3y=1$

Ans. CD

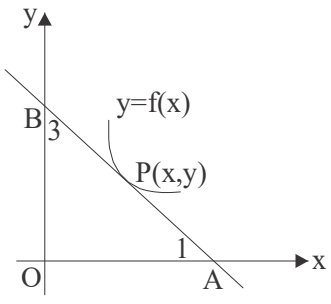
Sol. Equation of the tangent to the curve $y=f(x)$ at (x,y) is

$$Y-y=\frac{dy}{dx}(X-x)$$

$\because$ Thus, cuts the x–axis at A and y–axis at B.

$$\therefore A\left(\frac{x\frac{dy}{dx}-y}{\frac{dy}{dx}},0\right) \text{ and } B\left(0,-x\frac{dy}{dx}+y\right)$$

$\because BP:PA=3:1$



$$\Rightarrow x=\frac{3\left(x\frac{dy}{dx}-y\right)}{(dy/dx)}+1\times 0}{4}$$

$$\Rightarrow x\frac{dy}{dx}+3y=0$$

$\Rightarrow \int \frac{dy}{y} = \int -3\frac{dx}{x}$

$\Rightarrow \log y = -3\log x + \log C$

$\Rightarrow y = \frac{C}{x^3}$

$\because$ Curve passes through $(1,1) \therefore C = 1$

$\therefore$ Curve is $x^3y = 1$ which passes through $\left(3, \frac{1}{27}\right)$

SECTION–IV : (Maximum Marks : 16)

- This section contains **TWO** questions.
- Each question contains two columns, **Column-I** and **Column-II.**
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **column-II.**
- One or more entries in **Column-I** may match with one or more entries in **Column-II.**
- The ORS contains a 4 × 5 matrix whose layout will be similar to the one shown below :

(A) (P) (Q) (R) (S) (T)
(B) (P) (Q) (R) (S) (T)
(C) (P) (Q) (R) (S) (T)
(D) (P) (Q) (R) (S) (T)

- For each entry in **column-I,** darken the bubbles of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then darken these three bubbles in the ORS. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks	: +2	If only the bubble(s) corresponding to all the correct match(es) is (are) darkened
Zero Marks	: 0	In none of the bubbles is darkened
Negative Marks	: 0	In all other cases

9. Let $a < b < c < d$, match with the column in which equation has root in:

Column I	**Column II**
(A) $(x-a)(x-c)+2007(x-b)(x-d)=0$	(P) (c,d)
(B) $2007(x-a)(x-c)-2006(x-b)(x-d)=0$	(Q) (a,b)
(C) $(x-a)(x-b)(x-c)+(x-d)=0$	(R) $(-\infty, a)$
(D) $(x-a)(x-b)+(x-c)=0$	(S) (b,c)

Ans. $A:(P,Q), B:(R,S), C(P), D(R,S)$

Sol. (A) $(x-a)(x-c)+2007(x-b)(x-d)=0$

$f(a) = 2007(a-b)(a-d) > 0$

$f(b) = (b-a)(b-c) < 0$

$f(c) = 2007(c-b)(c-d) < 0$

$f(d)=(d-a)(d-c)>0$

$\Rightarrow$ Root lies in (a,b) and (c,d)

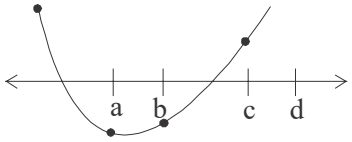
(B) $2007(x-a)(x-c)-2006(x-b)(x-d)=0$

$f(a)=-2006(a-b)(a-d)<0$

$f(b)=2007(b-a)(b-c)<0$

$f(c)=-2006(c-b)(c-d)>0$

$f(d)=2007(d-a)(d-c)>0$



Roots lies in $(-\infty,a)\&(b,c)$

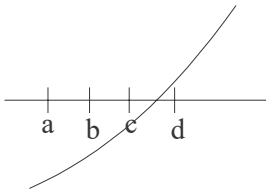
(C) $(x-a)(x-b)(x-c)+(x-d)=0$

$f(a)=a-d<0$

$f(b)=b-d<0$

$f(c)=(c-d)<0$

$f(d)=(d-a)(d-b)(d-c)>0$

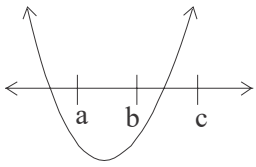


(D) $(x-a)(x-b)+(x-c)=0$

$f(a)=a-c<0$

$f(b)=b-c<0$

$f(c)=(c-a)(c-b)>0$



Root lies in $(-\infty,a)\&(b,c)$

10. Match curves with their Differential Equations:

Column I	**Column II**
(A) $y=c_1e^{2x}+c_2e^{3x}+c_3e^{-x}$	(P) $\frac{d^3y}{dx^3}-6\frac{d^2y}{dx^2}+12\frac{dy}{dx}-8y=0$
(B) $y=(c_1+c_2x+c_3x^2)e^{2x}$	(Q) $x\left[y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2\right]=y\frac{dy}{dx}$

(C) $xy = ae^{x} + be^{-x} + x^{2}$

(R) $\frac{d^3y}{dx^3} - 4\frac{d^2y}{dx^2} + \frac{dy}{dx} + 6y = 0$

(D) $ax^2 + by^2 = 1$

(S) $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - xy + x^2 - 2 = 0$

Ans. A : R, B : P, C : S, D : Q

Sol. (A) $y = c_1e^{2x} + c_2e^{3x} + c_3e^{-x}$

$y' = 2c_1e^{2x} + 3c_2e^{3x} - c_3e^{-x}$

$y'' = 4c_1e^{2x} + 9c_2e^{3x} + c_3e^{-x}$

$y''' = 8c_1e^{2x} + 27c_2e^{3x} - c_3e^{-x}$

$y''' - 4y'' + y' + 6y = 0$

(B) $y = \left(c_1 + c_2x + c_3x^2\right)e^{2x}$

$ye^{-2x} = c_1 + c_2x + c_3x^2$

$y'e^{-2x} - 2ye^{-2x} = c_2 + 2c_3x$

$y''e^{-2x} - 4y'e^{-2x} + 4y^{-2x} = 2c_3$

$\left(y''' - 6y'' + 12y' - 8\right)e^{-2x} = 0$

(C) $xy = ae^{x} + be^{-x} + x^{2}$

$xy' + y = ae^{x} - be^{-x} + 2x$

$xy'' + y' + y' = ae^{x} + be^{-x} + 2$

$xy'' + 2y' = xy - x^2 + 2$

$xy'' + 2y' - xy + x^2 - 2 = 0$

(D) $ax^2 + by^2 = 1$

$ax + byy' = 0$

$\Rightarrow \frac{a}{b} + yy' = 0$

$x\left(y'^2 + yy''\right) - yy' = 0$

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :

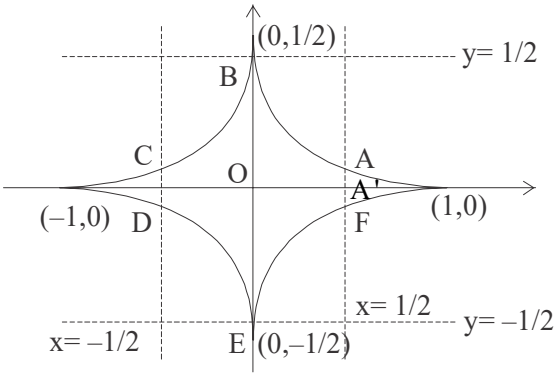
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

11. The area of the region bounded by the curves $2^{|x|}|y| + 2^{|x|-1} \le 1, |x| \le \frac{1}{2}$ and $|y| \le \frac{1}{2}$. is

$\frac{a}{\log 2}\left(1-\frac{1}{\sqrt{b}}\right)+c$ then $a+b+c$ is $(a,b,c\in N)$

Ans. 5

Sol.



$2^{|x|}|y|+2^{|x|-1}\le 1$ is equal to $|y|\le 2^{-|x|}-\frac{1}{2}$

The curve is symmetric about both the x and y–axis.

Required region is shaded ABCDEA

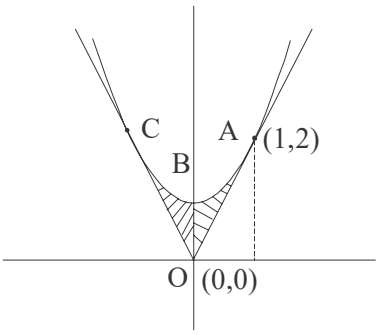
Required area = 4 area of the region OA'ABO

$$=4\int_0^{1/2}\left(2^{-x}-\frac{1}{2}\right)dx=\frac{4}{\log 2}\left(1-\frac{1}{\sqrt{2}}\right)-1$$

12. The area bounded by $y=x^2+1$ and the tangents to it drawn from the origin is A then 3A is

Ans. 2

Sol. The parabola is even function & let the equation of tangent is $y=mx$



Now we calculate the point of intersection of parabola & tangent $mx=x^2+1$

$x^2-mx+1=0\Rightarrow D=0$

$\Rightarrow m^2-4=0\Rightarrow m=\pm 2$

Two tangents are possible $y=2x$ & $y=-2x$

Intersection of $y=x^2+1$ & $y=2x$ is

$x=1$ & $y=2$

Area of shaded region

$$OAB=\int_0^1(y_2-y_1)dx=\int_0^1\left((x^2+1)-2x\right)dx$$

$= \frac{1}{3}$ sq.units

Area of total shaded region $= 2\left(\frac{1}{3}\right) = \frac{2}{3}$ sq.units

13. If $\int_{-5}^{-2}\left(\frac{x^2-x}{x^3-3x+1}\right)^2 dx + \int_{\frac{1}{6}}^{\frac{1}{3}}\left(\frac{x^2-x}{x^3-3x+1}\right)^2 dx + \int_{\frac{6}{5}}^{\frac{3}{2}}\left(\frac{x^2-x}{x^3-3x+1}\right)^2 dx = \frac{p}{q}$, where p and q are co–primes, then $\frac{(6q-p)}{6}$ is

Ans. 5

Sol. $I_1 = \int_{-5}^{-2}\left(\frac{x^2-x}{x^3-3x+1}\right)^2 dx$

$I_2 = \int_{\frac{1}{6}}^{\frac{1}{3}}\left(\frac{x^2-x}{x^3-3x+1}\right)^2 dx$

Put $x = \frac{1}{(1-t)}$

$I_2 = \int_{-5}^{-2}\left(\frac{t^2-t}{t^3-3t+1}\right)\left(\frac{1}{1-t}dt\right)$

In I_3, put $x = 1-\frac{1}{u}$

$I_3 = \int_{-5}^{-2}\left(\frac{u^2-u}{u^3-3u+1}\right)^2 \frac{du}{u^2}$

$I_1 + I_2 + I_3 = \int_{-5}^{-2}\left(\frac{x^2-x}{x^2-3x+1}\right)^2\left(1+\frac{1}{(1-x)^2}+\frac{1}{x^2}\right)dx$

Let $v = x+1-\frac{1}{x}-\frac{1}{(x-1)}$

$dv = \left(1+\frac{1}{x^2}+\frac{1}{(x-1)^2}\right)dx$

$I_1 + I_2 + I_3 = \int_{-\frac{109}{30}}^{-\frac{1}{6}} \frac{dv}{vz} = \frac{624}{109}$

$p = 624,\ q = 109$

14. The area given by $x+y\le 6, x^2+y^2\le 6y$ and $y^2\le 8x$ is $\frac{a}{b}\pi-\frac{1}{6}$ then $a+b$ is (given $a,b\in N$ a, b are relatively prime)

Ans. 13

Sol. Let us consider the curves

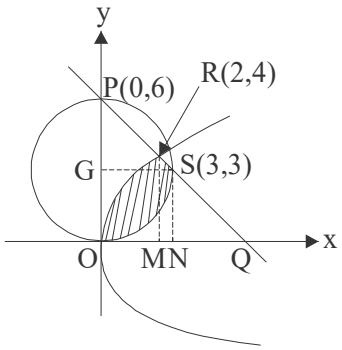
$P\equiv y^2-8x=0 \qquad -----(i)$

$C\equiv x^2+y^2=6y$

i.e., $x^2+(y-3)^2-9=0 \qquad ---(ii)$

and $S\equiv x+y-6 \qquad ---(iii)$

The intersection points of the curves (ii) and (iii) are given by



$(6-y)^2+y^2-6y=0$

i.e., $y=3,6$

Therefore, the points are $(0,6)$ and $(3,3)$. The intersection points of the curves (i) and (iii) are given by

$y^2=8(6-y)$, ie, $y=4,-12$

Therefore, the point of intersection in 1st quadrant is (2,4).

Now, we know that

$C\le 0$ denotes the region, inside the circle C=0

$P\le 0$ denotes the region, inside the parabola P=0

$S\le 0$ denotes the region, which is (–ve) side of the line S=0

$\therefore$ The required area = Area of curvilinear $\Delta OMRO$

+ Area of trapezium MNSR – Area of curvilinear $\Delta ONSO$

$=\int_0^2\sqrt{8x}\,dx+\frac{1}{2}(MR+NS)\cdot MN$ – (Area of square ONSG –Area of sector OSGO)

$=\int_0^2\sqrt{8x}\,dx+\frac{1}{2}(4+3)\cdot 1-\left(3^2-\frac{\pi\cdot 3^2}{4}\right)$

$=\frac{9\pi}{4}-\frac{1}{6}$

15. Let $P(x)=x^6-x^5-x^3-x^2-x$ and $\alpha,\beta,\gamma,\delta$ are the roots of the equation $x^4-x^3-x^2-1=0$, then $P(\alpha)+P(\beta)+P(\gamma)+P(\delta)=$

Ans. 6

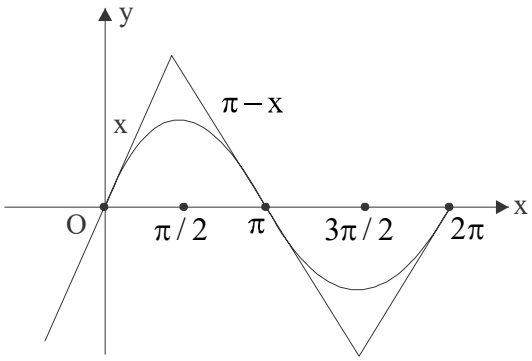
Sol. $P(x)=(x^4-x^3-x^2-1)(x^2+)+(x^2-x+1)$

$P(\alpha)+P(\beta)+P(\gamma)+P(\delta)=(\alpha^2-\alpha+1)+(\beta^2-\beta+1)+(\gamma^2-\gamma+1)+(\delta^2-\delta+1)=6$

16. $\int_0^{2\pi}\max\{\sin x,\sin^{-1}(\sin x)\}dx=\frac{\pi^b}{a}-c$, then $(a+b+c)$ is $(a,b,c\in N)$

Ans. 8

Sol.



$I=\int_0^{2\pi}\max(\sin x,\sin^{-1}(\sin x))dx$

$=\int_0^{\pi/2}x\,dx+\int_{\pi/2}^{\pi}(\pi-x)dx+\int_{\pi}^{2\pi}(\sin x)dx$

$=\frac{(\pi^2-8)}{4}$

17. If real roots of $(x+3)^4+(x+5)^4=16$ are a & b then $\frac{a^2+b^2+ab}{7}$ is

Ans. 7

Sol. Put $y=x+\frac{3+5}{2}=x+4$

The equation (1) becomes $(y-1)^4+(y+1)^4=16$

$\Rightarrow 2[y^4+6y^2+1]=16\Rightarrow y^4+6y^2-7=0$

$\Rightarrow(y^2+7)(y^2-1)=0\rightarrow y^2=-7$ or $y^2=1$

$\Rightarrow y=\pm\sqrt{7}\,i$ or $y=\pm1\Rightarrow x=-4\pm\sqrt{7}\,i$ or $x=-4\pm1=-5$ or -3

Thus, the given equation has two real roots.

18. The number of integral values of a for which the inequality $3-|x-a|>x^2$ is satisfied by at least one negative x, must be equal to

Ans. 6

Sol. Given inequality is $3-|x-a|>x^2$ ----(i)

$\Rightarrow|x-a|<3-x^2$

Or $-3+x^2<x-a<(3-x^2)$

Or $x^2 - x + a - 3 < 0 \quad ----(ii)$

And $x^2 + x - (a+3) < 0 \quad ----(iii)$

If $\alpha, \beta (\alpha < \beta)$ be the real and unequal roots of the corresponding, equation of eq. (i) for all $x \in (\alpha, \beta)$. Since eq.(i) will have at least one negative solution. The corresponding equation of (ii) and (iii) are

$x^2 - x + a - 3 = 0 \quad ----(iv)$

And $x^2 - x(a+3) = 0 \quad ----(v)$

Respectively

Both eqs. (iv) and (v) should have real and unequal roots, then

$1 - 4.1.(a-3) > 0$ and $1 + 4.(a+3) > 0$

i.e. $a < \frac{13}{4}$ and $a > -\frac{13}{4} \quad ----(vi)$

Roots of eq. (iv) are $\frac{1 \pm \sqrt{(13-4a)}}{2}$

One of these will be negative provided $13 - 4a > 1$

i.e. $a < 3$

And roots of eq. (v) are $-\frac{1 \pm \sqrt{(13+4a)}}{2}$

So, it has one negative root.

Combining eqs. (vi) and (vii), we get

$-\frac{13}{4} < a < 3$

$\therefore$ Integral values of a are $-3, -2, -1, 0, 1, 2.$

Total number = 6